

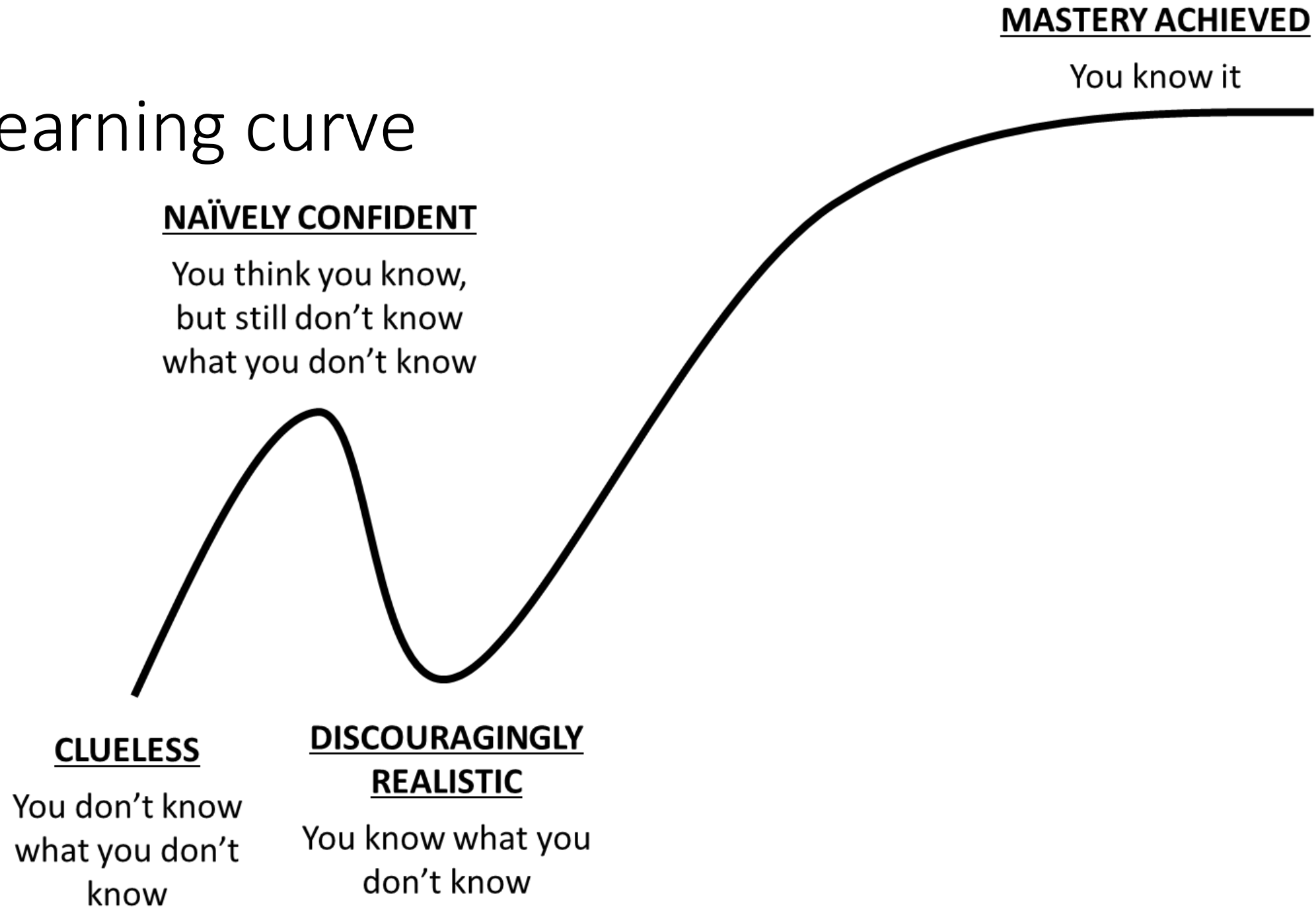
Statistics with Spa ows II

Many models, matrices, and magic

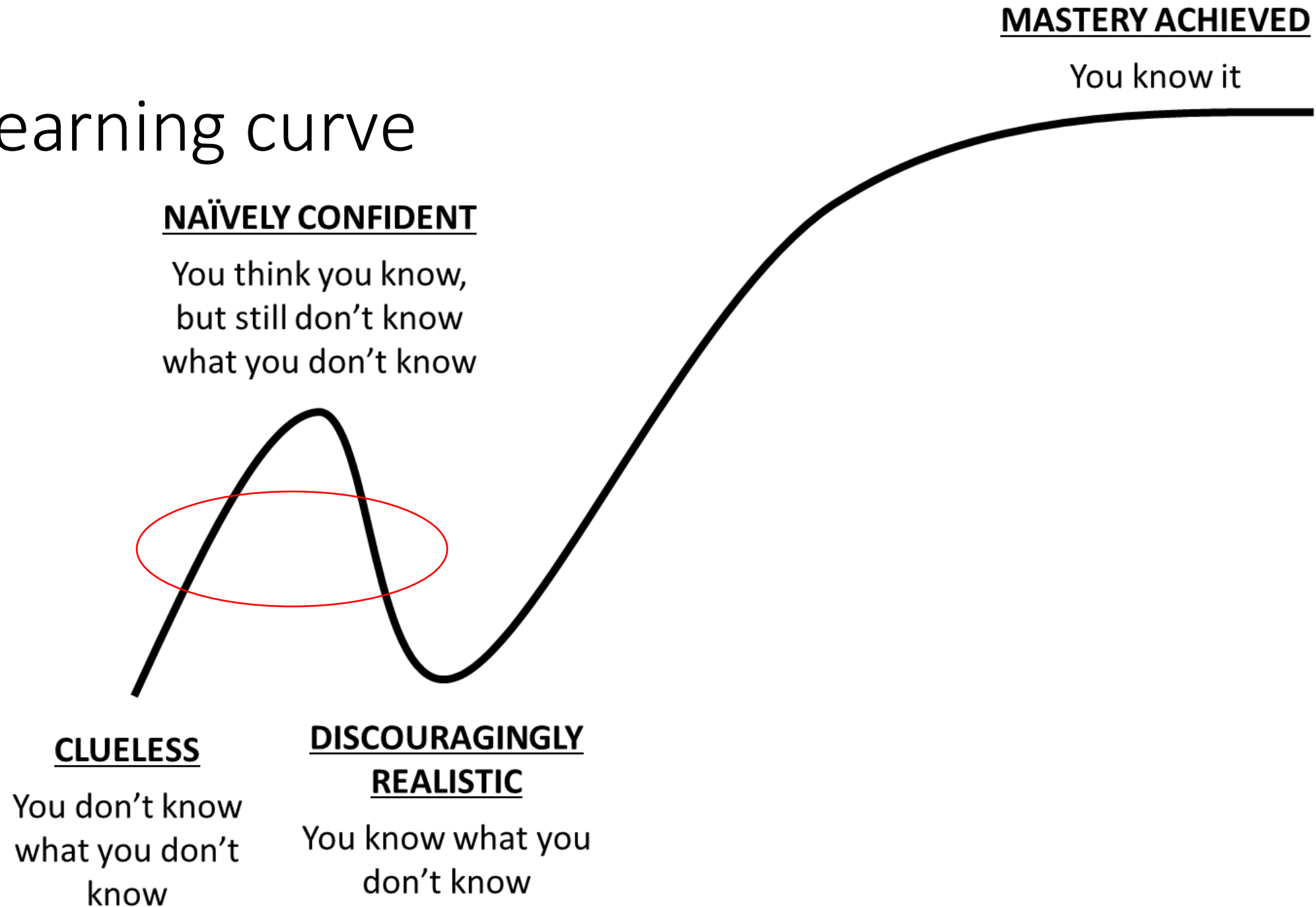
Julia Schroeder

Julia.schroeder@imperial.ac.uk

Learning curve



Learning curve



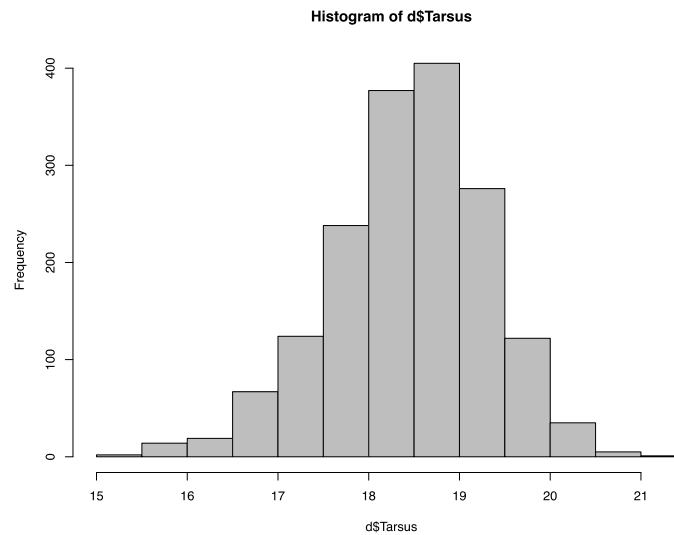
Learning curve



I never make the same
mistake twice. I make it
like five or six time, you
know, just to be sure.

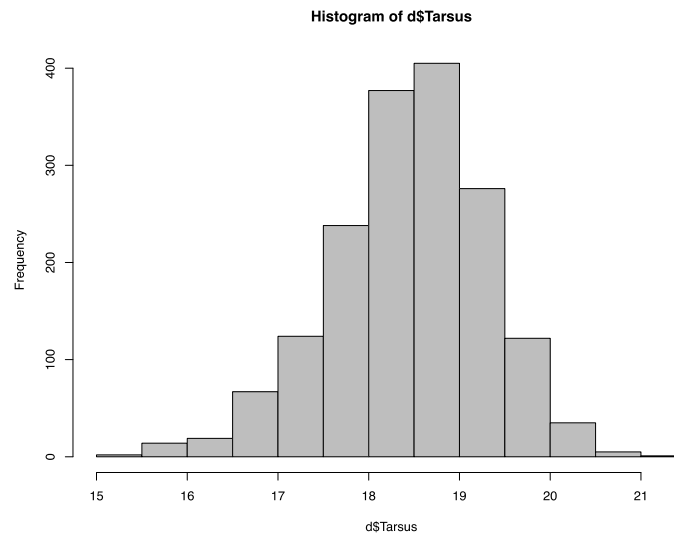


Describing data distributions



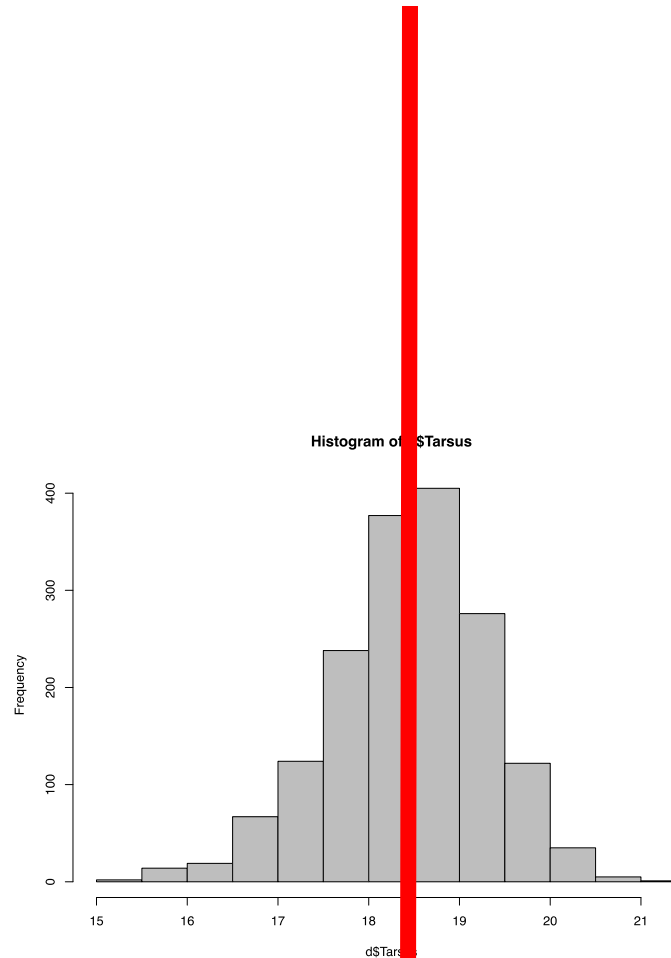
Describing data distributions

- Centrality



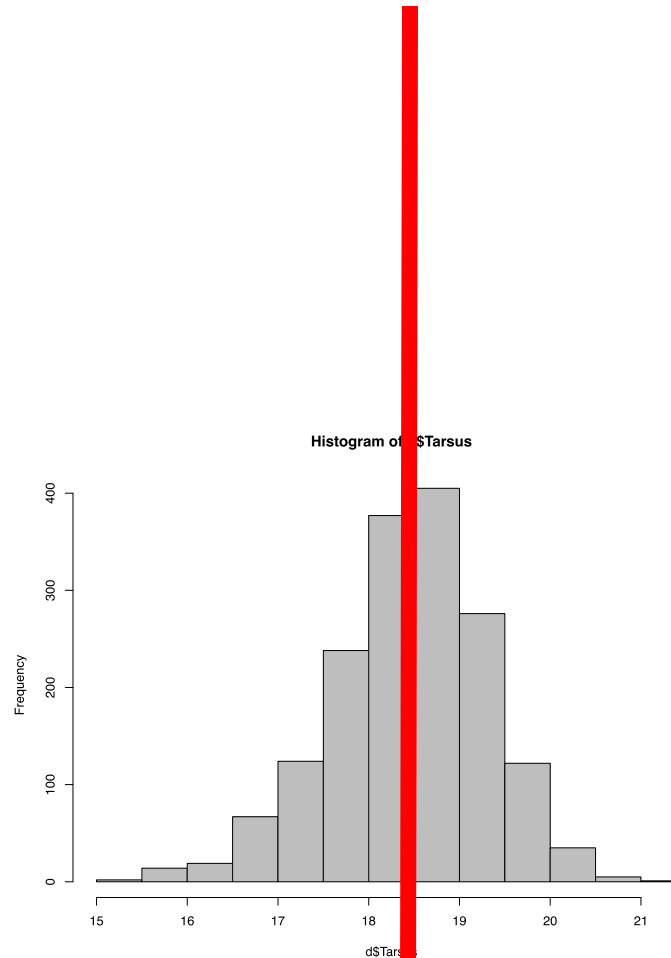
Describing data distributions

- Centrality



Describing data distributions

- Centrality
 - Mean $\bar{\mu}$
 - Mode
 - Median

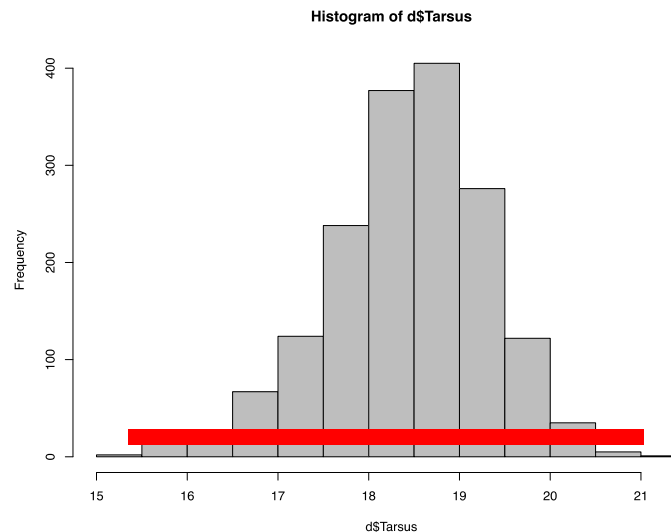


Describing data distributions

- Centrality

- Mean $\bar{\mu}$
- Mode
- Median

- Spread



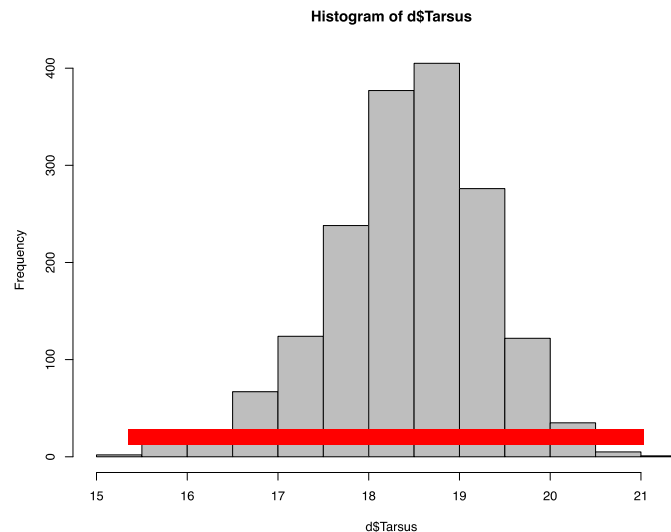
Describing data distributions

- Centrality

- Mean $\bar{\mu}$
- Mode
- Median

- Spread

- Range (min, max)



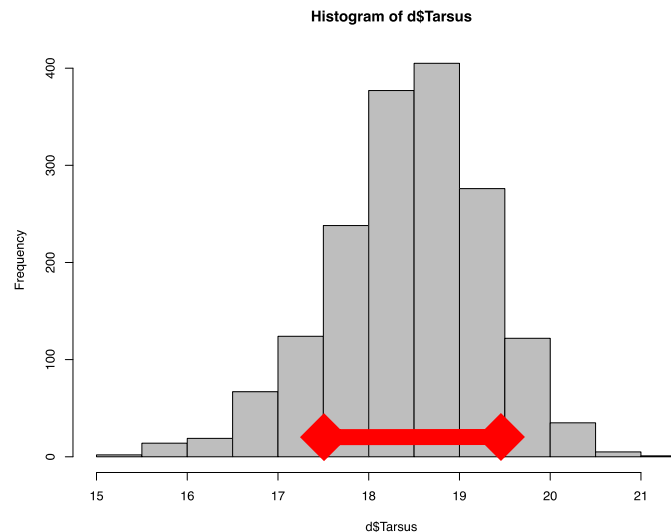
Describing data distributions

- Centrality

- Mean $\bar{\mu}$
- Mode
- Median

- Spread

- Range (min, max)
- Standard deviation σ



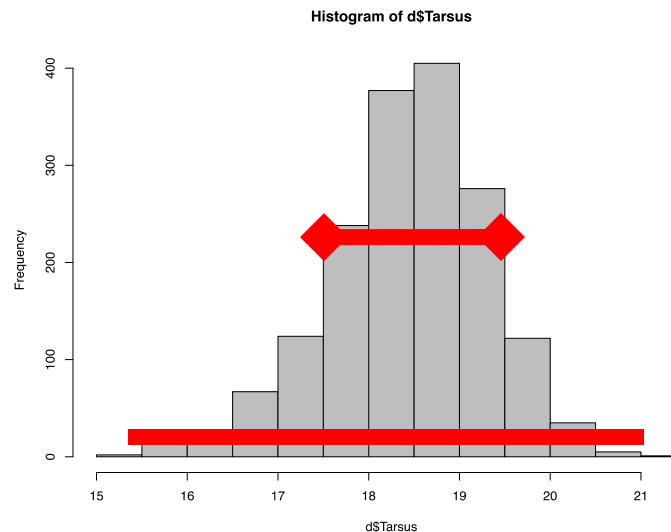
Describing data distributions

- Centrality

- Mean $\bar{\mu}$
- Mode
- Median

- Spread

- Range (min, max)
- Standard deviation σ
- Variance σ^2



Adding variances

$$\sigma_{Tarsus+Wing}^2 = \sigma_{Tarsus}^2 + \sigma_{Wing}^2 + 2COV_{Wing,Tarsus}$$

Adding variances

$$\sigma_{Tarsus+Wing}^2 = \sigma_{Tarsus}^2 + \sigma_{Wing}^2 + 2COV_{Wing,Tarsus}$$

$$\sigma_{Tarsus+Wing}^2 = \sigma_{Tarsus}^2 + \sigma_{Wing}^2$$

If *Tarsus* and *Wing* are independent variables

Linear models

- Basic building block of statistics
- Super useful
- Versatile
- Expandable

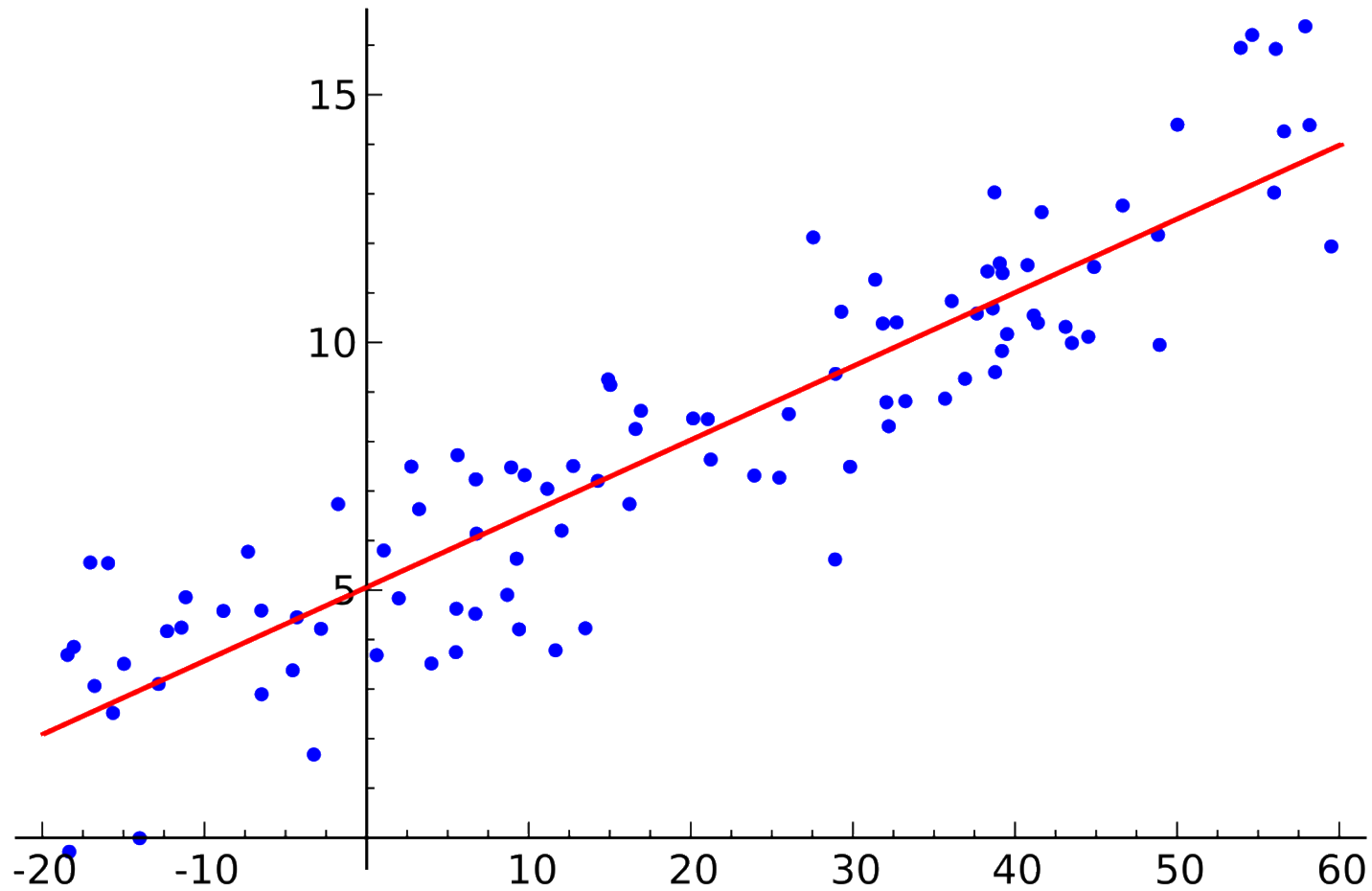
Fitting models to data



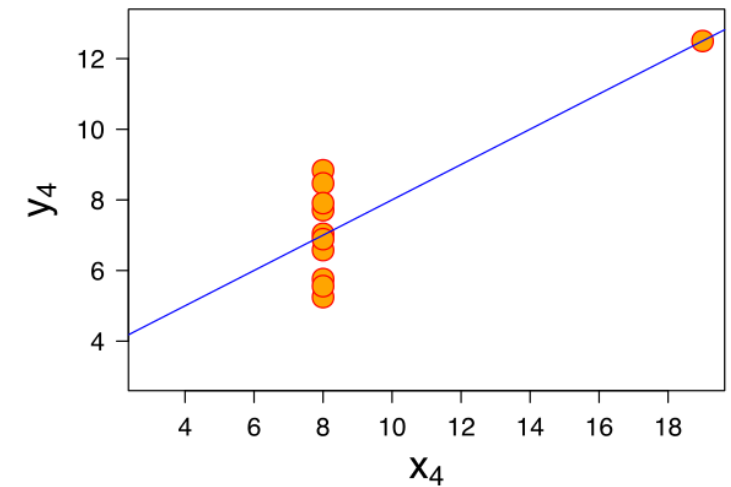
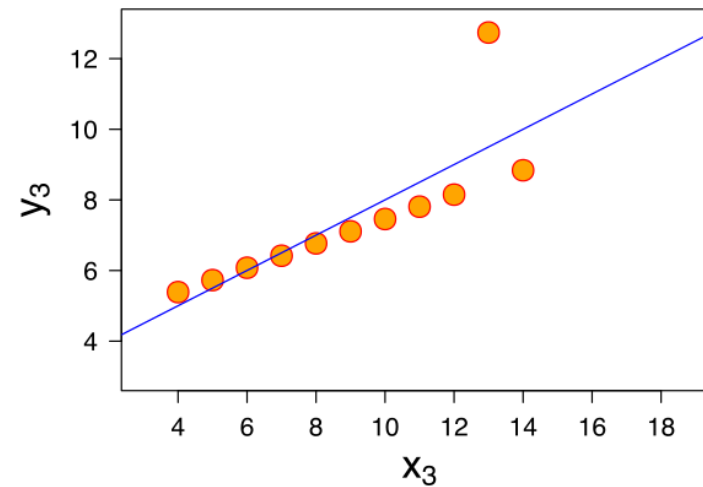
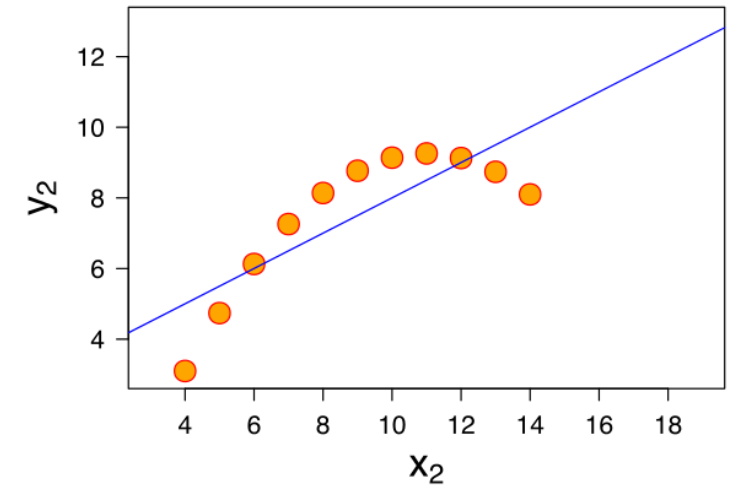
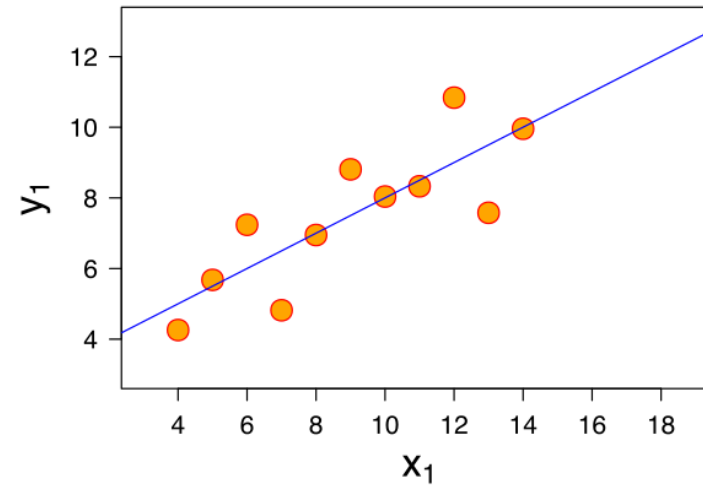
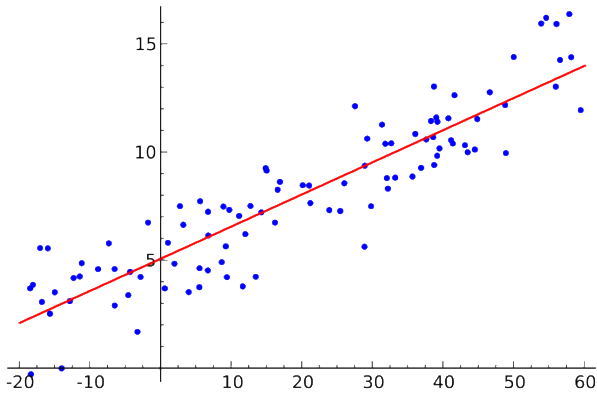
?



Fitting models to data



Fitting models to data

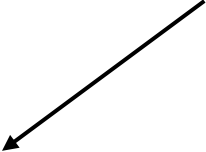


Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

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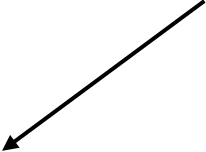


y
5
3
6
10
4

Data. Response variable,
e.g. sparrow body mass.

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$



y	i
5	1
3	2
6	3
10	4
4	5

Data. Response variable. Observation 1, 2, 3, etc.
e.g. sparrow body mass.

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

y i
5 1
3 2
6 3
10 4
4 5

Data. Response variable. Observation 1, 2, 3, etc.
e.g. sparrow body mass.

x i
3 1
1 2
4 3
8 4
2 5

Data. Explanatory variable.
e.g. sparrow tarsus length.

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$



ε	i
?	1
?	2
?	3
?	4
?	5

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$



$b_1 = ?$

ε	i
?	1
?	2
?	3
?	4
?	5

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$



$b_0 = ?$



$b_1 = ?$

ε	i
?	1
?	2
?	3
?	4
?	5

Linear models

- *Vectors* vs **scalars**

$$y_i = \mathbf{b}_0 + \mathbf{b}_1 x_i + \varepsilon_i$$

y i
5 1
3 2
6 3
10 4
4 5

$b_0 = ?$

$b_1 = ?$

x	i	ε	i
3	1	?	1
1	2	?	2
4	3	?	3
8	4	?	4
2	5	?	5

Linear models

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

- Find solution: these parameter estimates (scalars) that minimise the left-over error residuals (vector)

Linear models

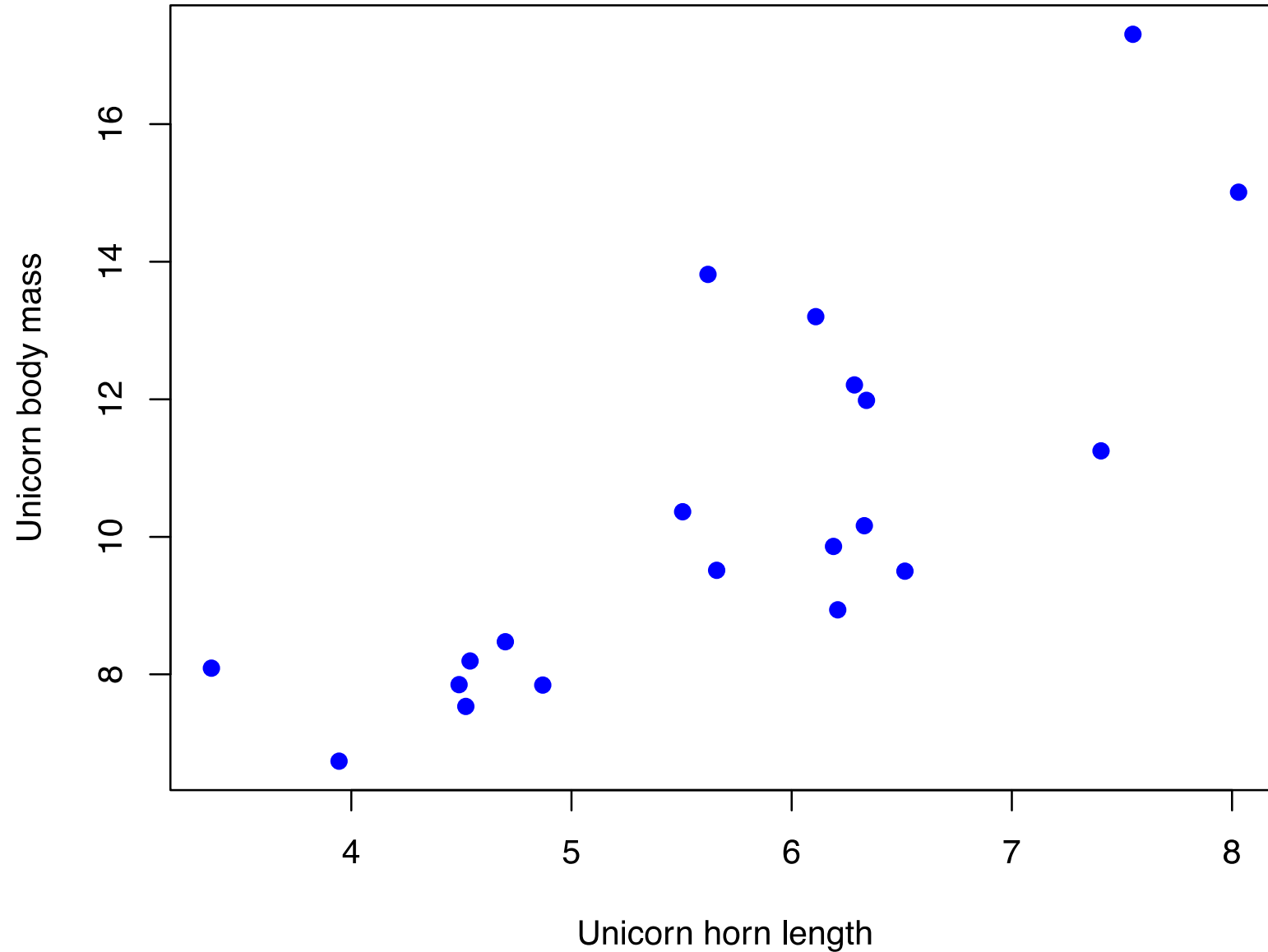
- Unicorn dataset – Runicorns.tx

```
> head(uni)
```

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3	Alice_Dogface_McDonald	Female	10.162390	6.330	0	8.190941	Spring	0
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5	Ratline_Slinger_Rose	Female	11.983053	6.340	0	4.287208	Spring	1
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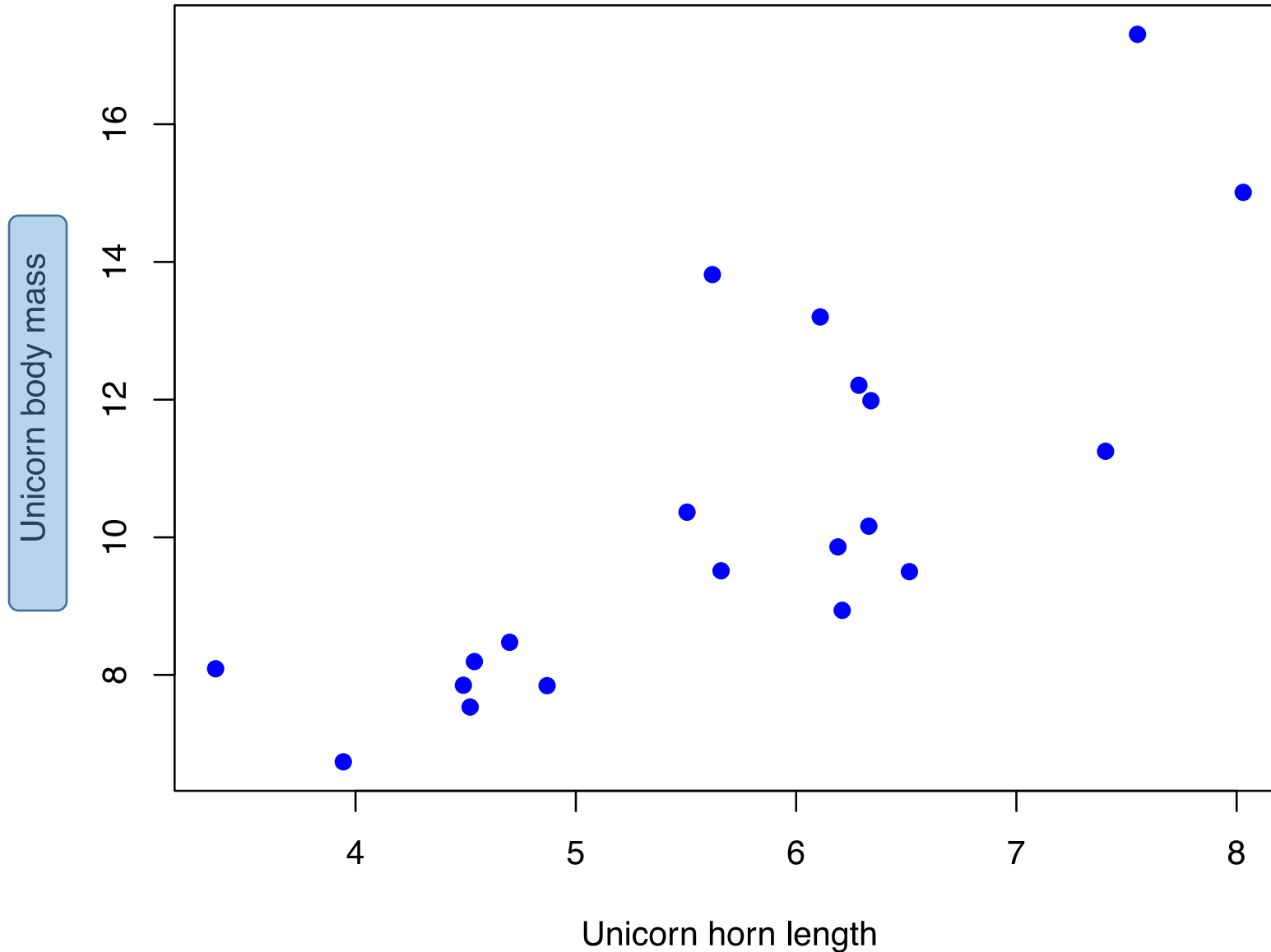
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> |
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Linear models - terminology



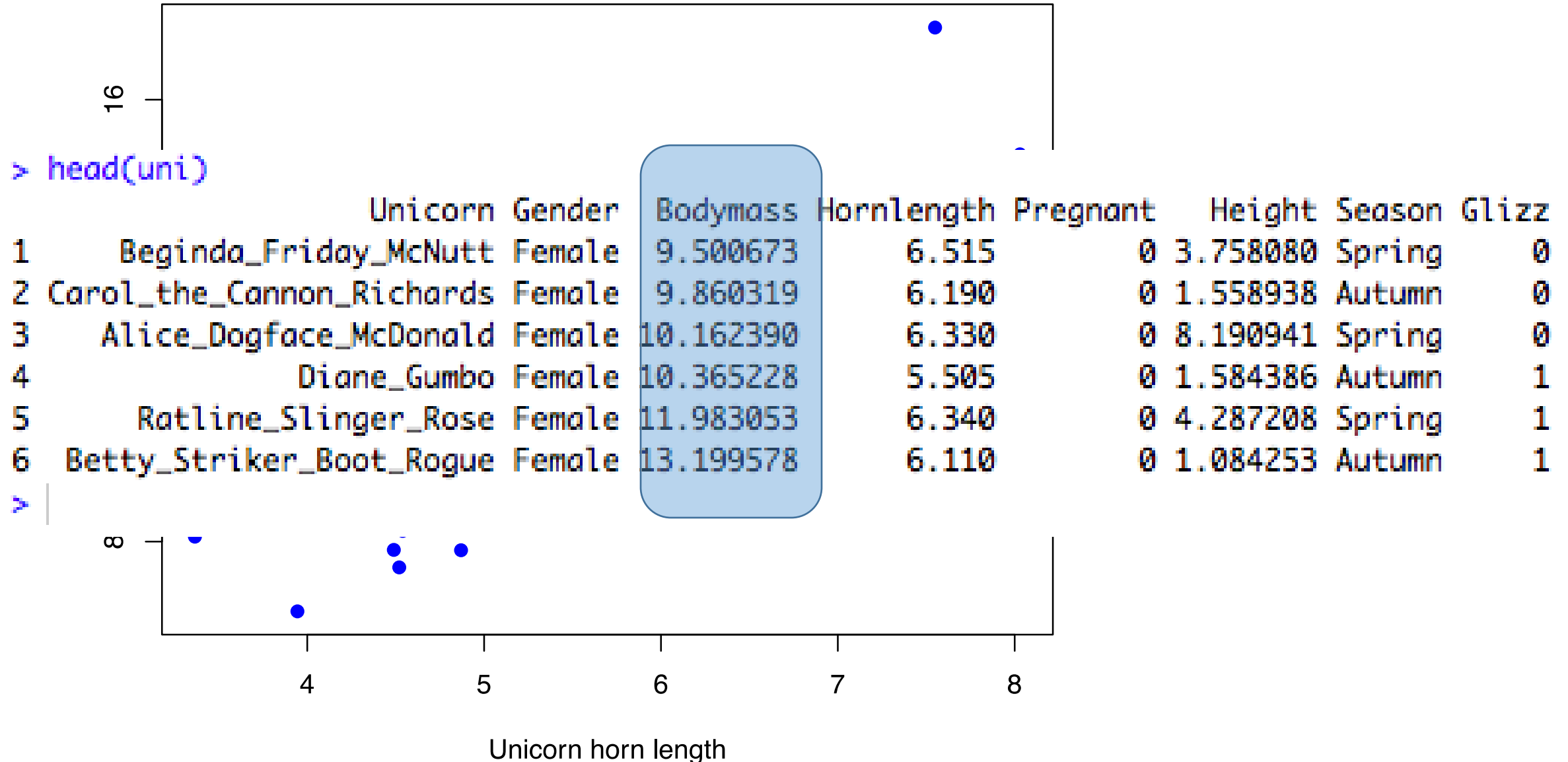
Linear models - terminology

Y = response variable, body mass



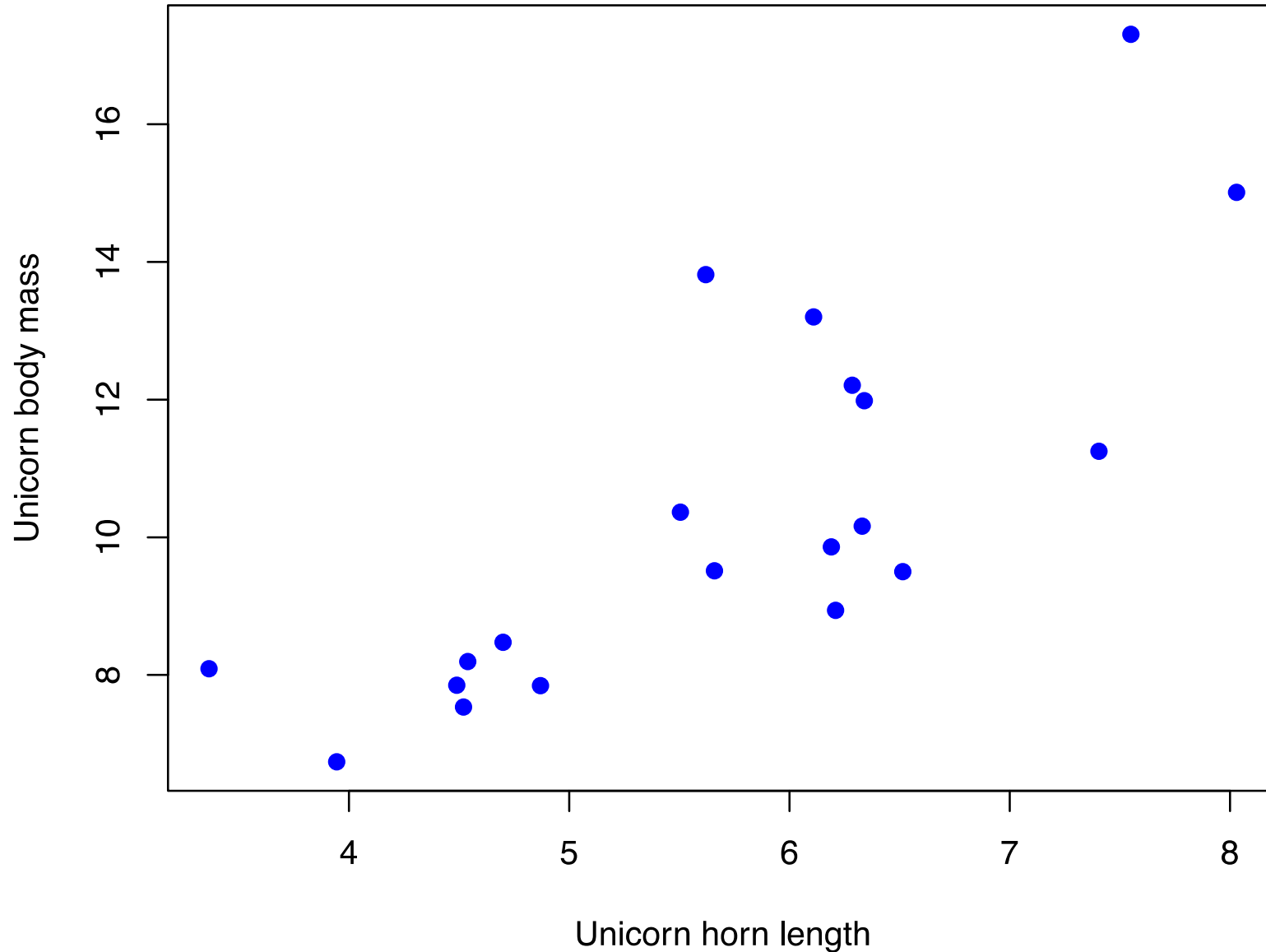
Linear models - terminology

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Linear models - terminology

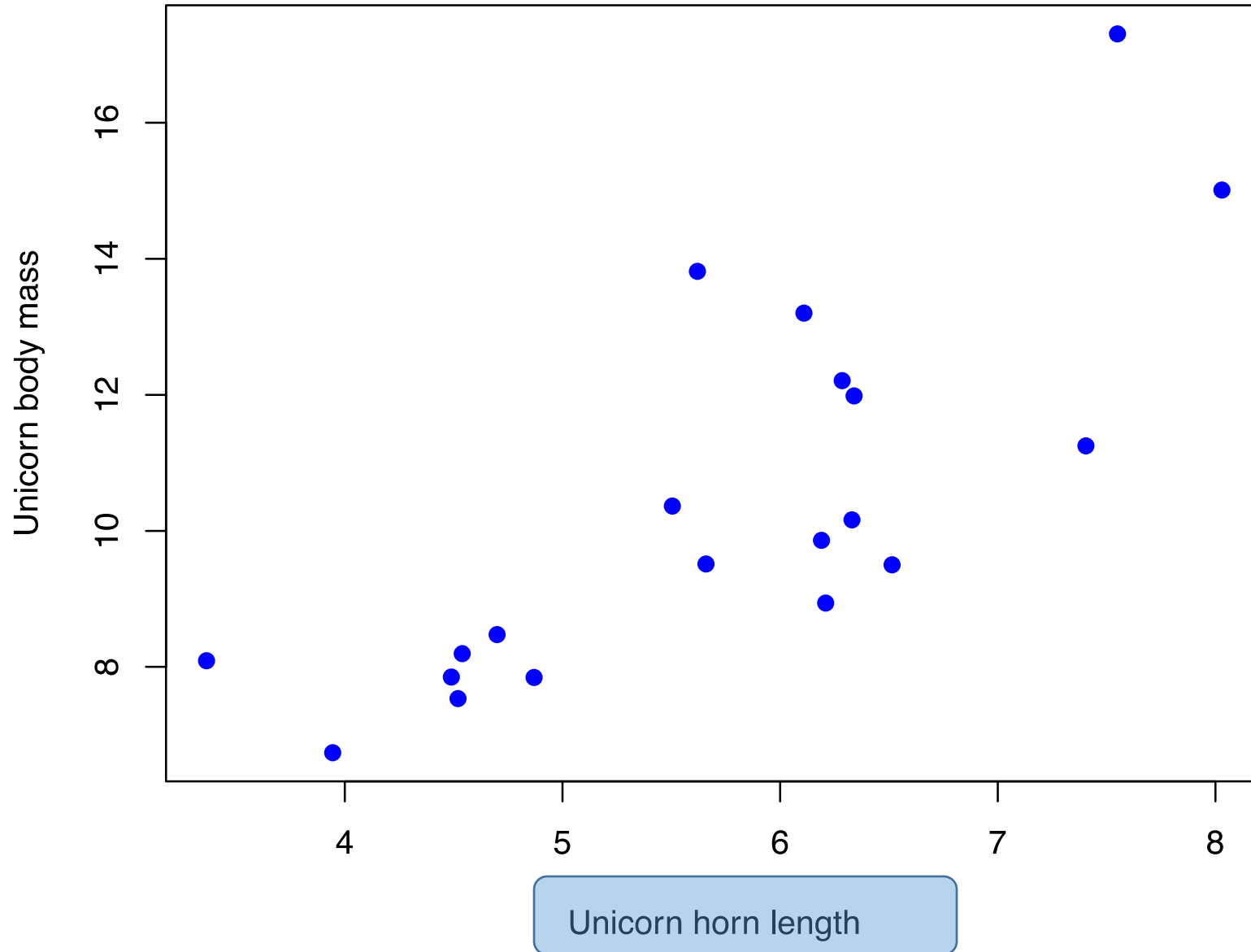
Y = response variable, body mass
X = explanatory variable, horn length



Linear models - terminology

Y = response variable, body mass

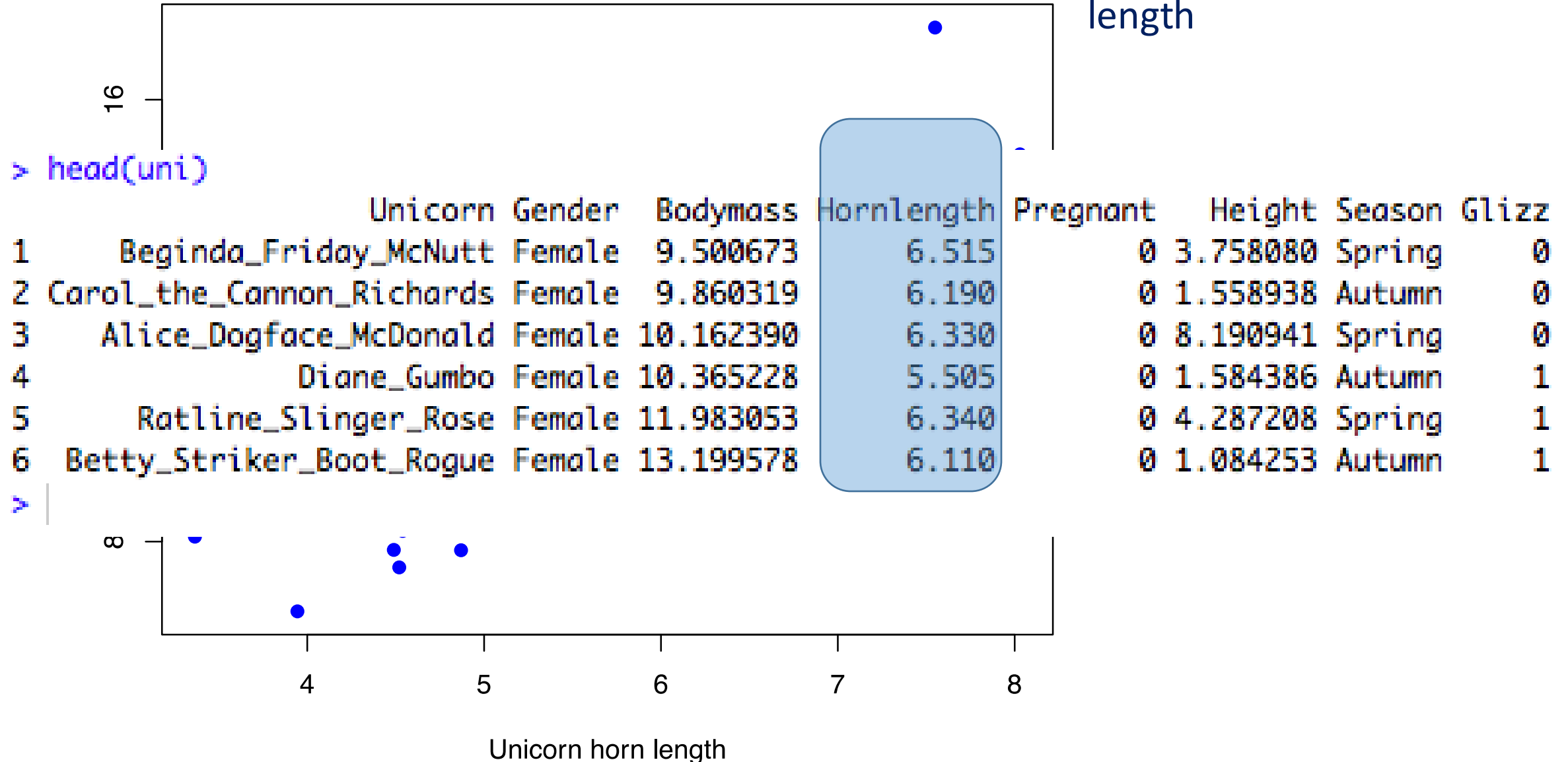
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Linear models - terminology

Y = response variable, body mass

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Linear models

- *Vectors* vs **scalars**

$$y_i = \mathbf{b}_0 + \mathbf{b}_1 x_i + \varepsilon_i$$

y i
5 1
3 2
6 3
10 4
4 5

$b_0 = ?$

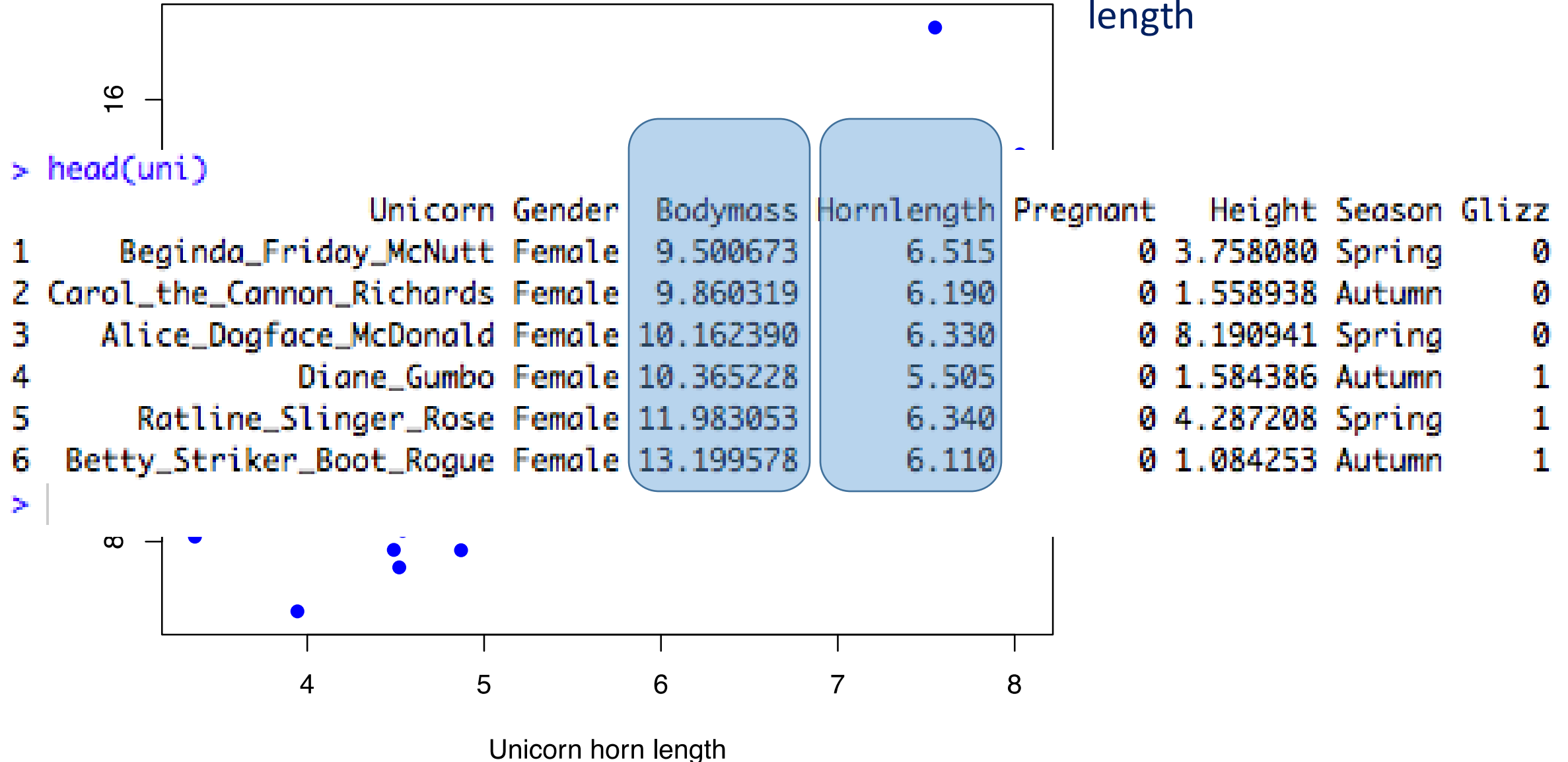
$b_1 = ?$

x	i	ε	i
3	1	?	1
1	2	?	2
4	3	?	3
8	4	?	4
2	5	?	5

Linear models - terminology

Y = response variable, body mass

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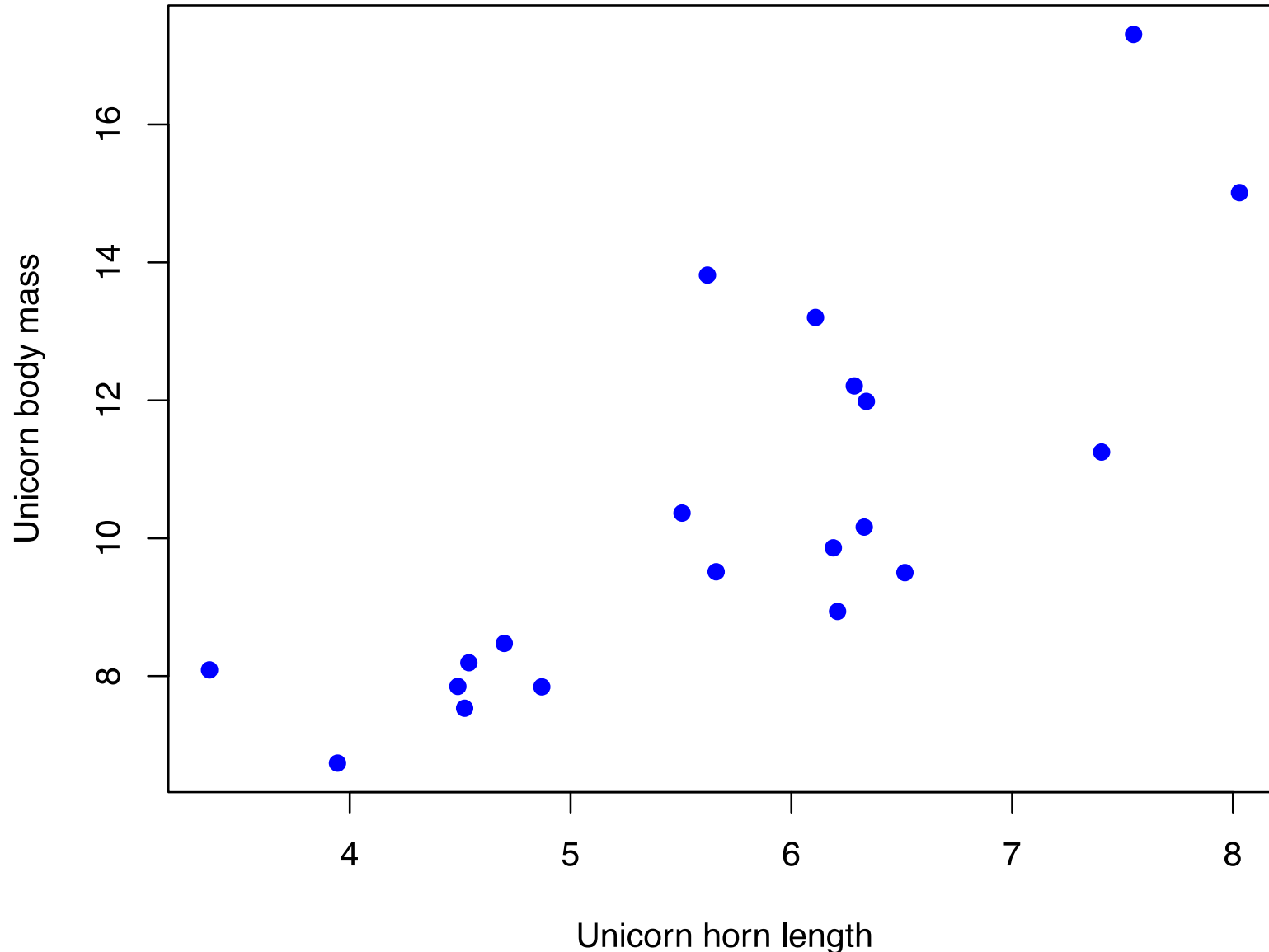


Linear models - terminology

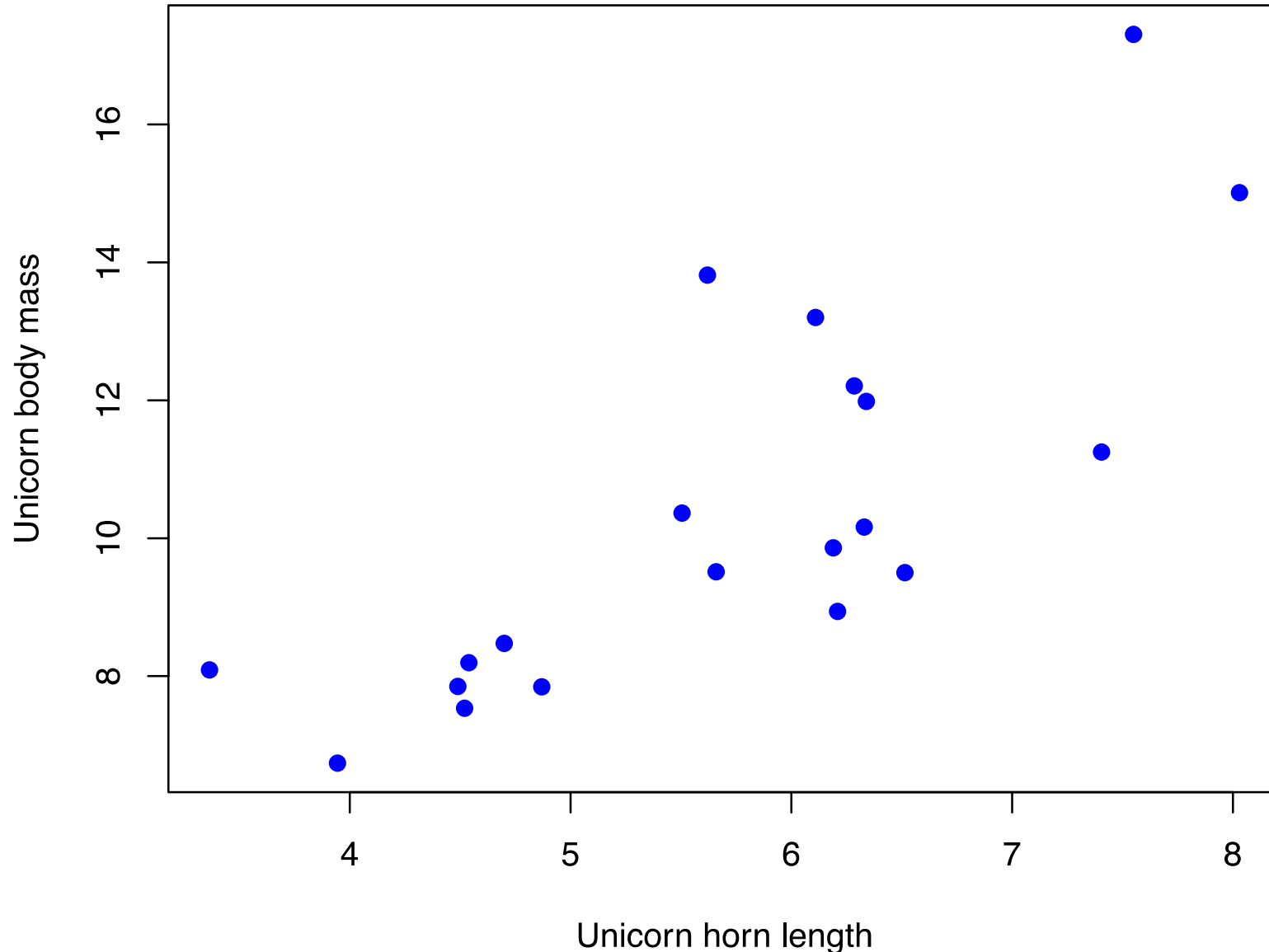
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X = explanatory variable, horn length

Y_i = OBSERVATIONS of body mass



Linear models - terminology



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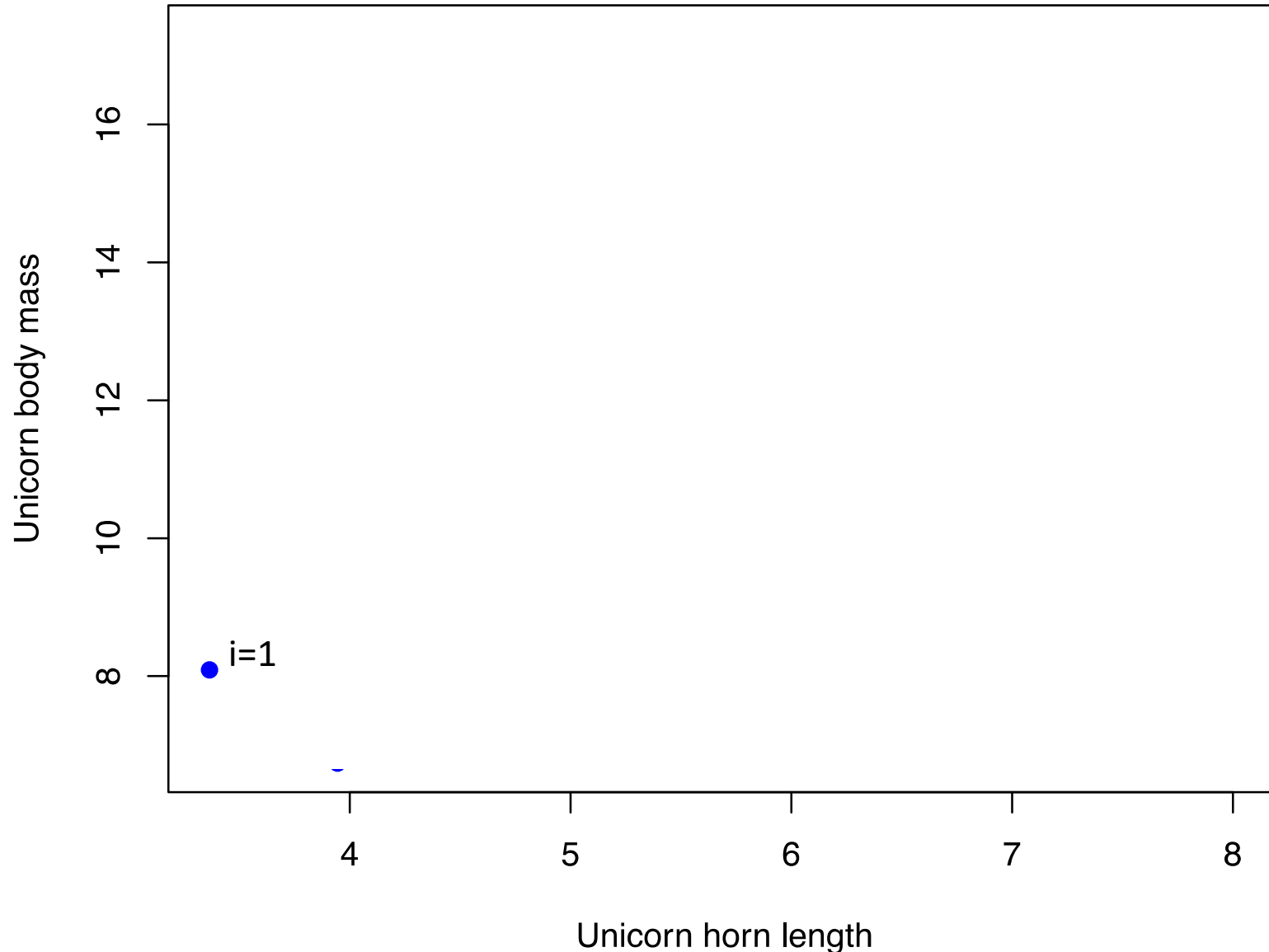
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```
> head(uni)
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4

5

6

7

8

Unicorn horn length

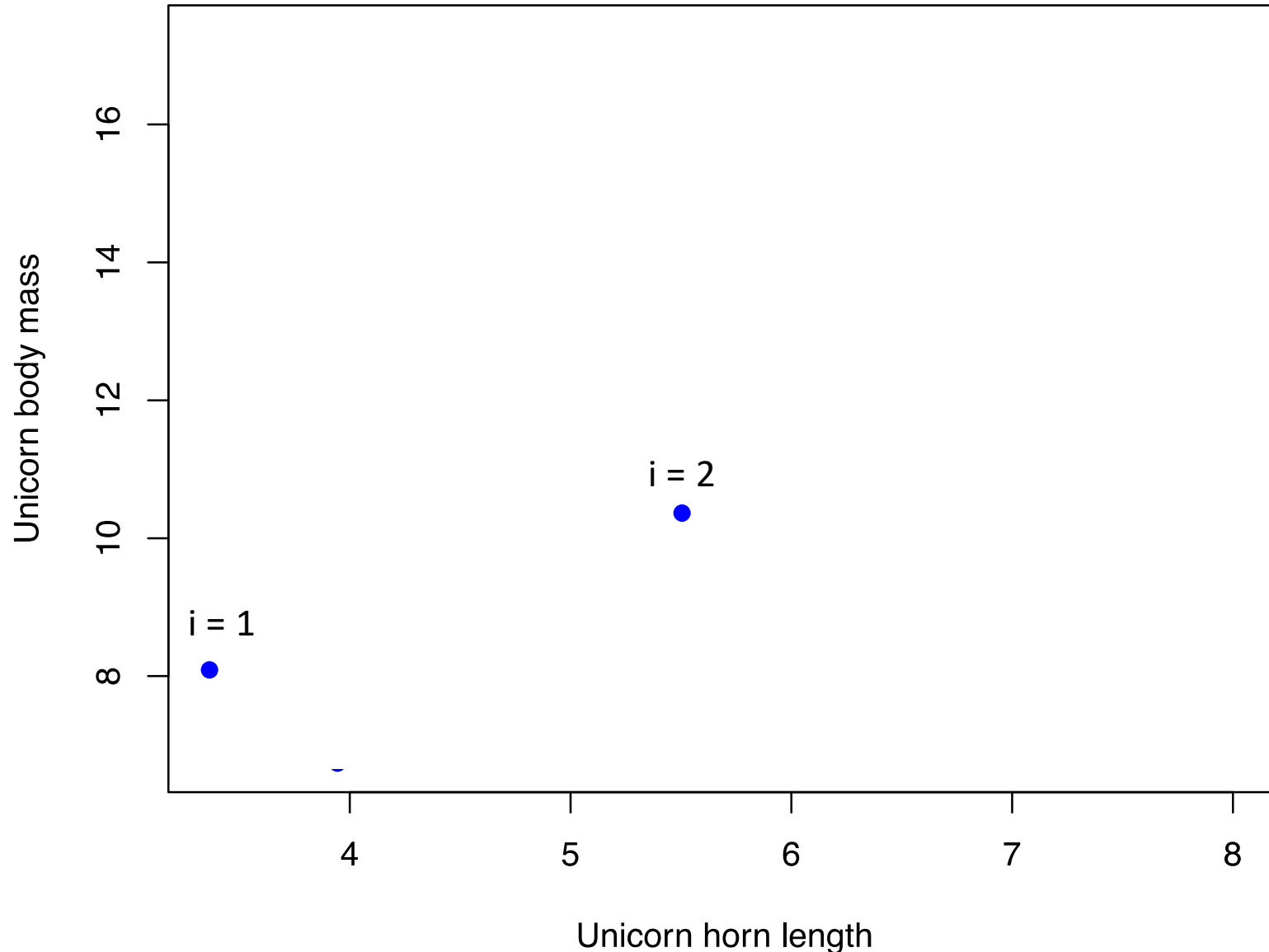
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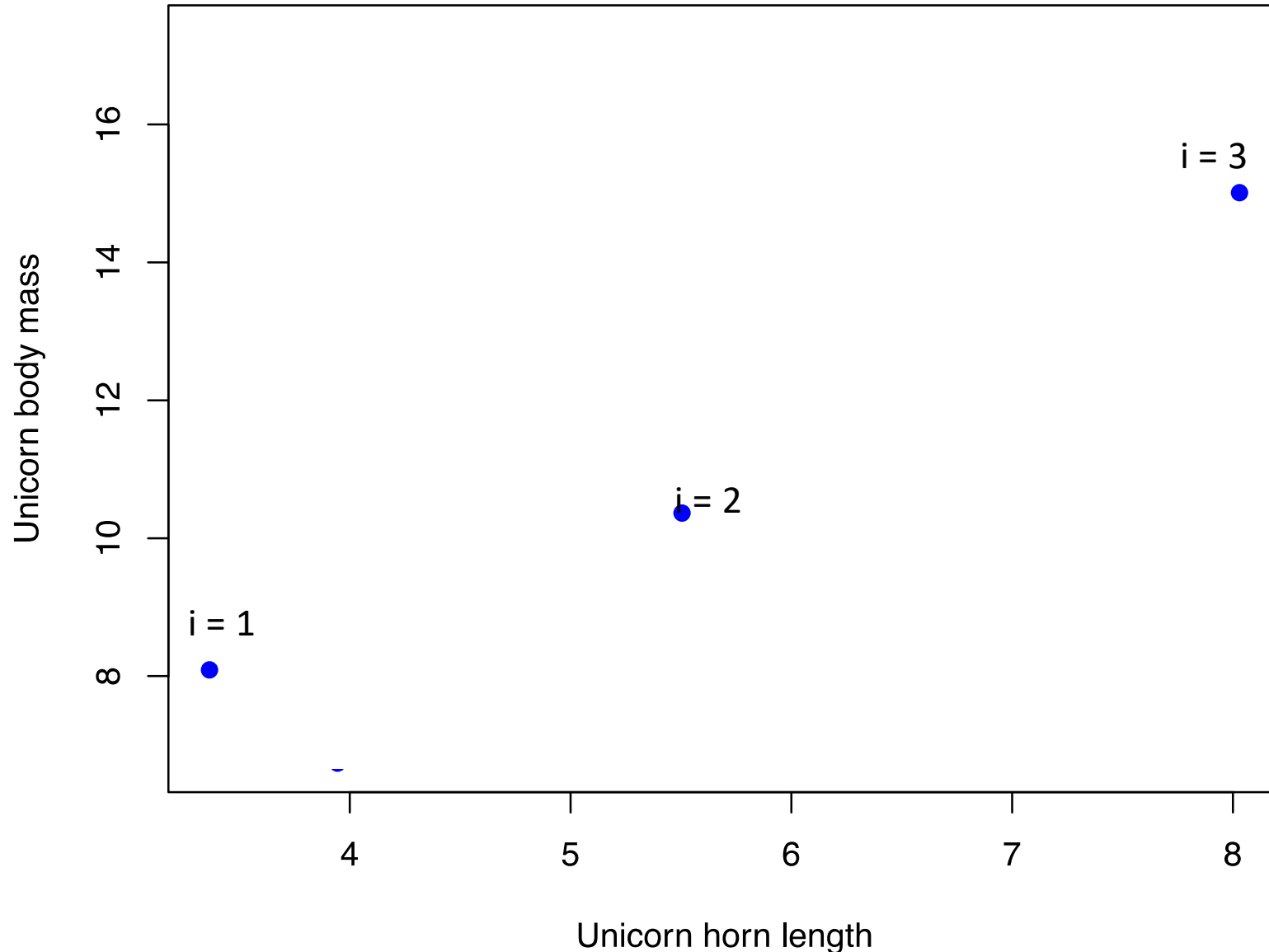
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Linear models - terminology

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Unicorn horn length

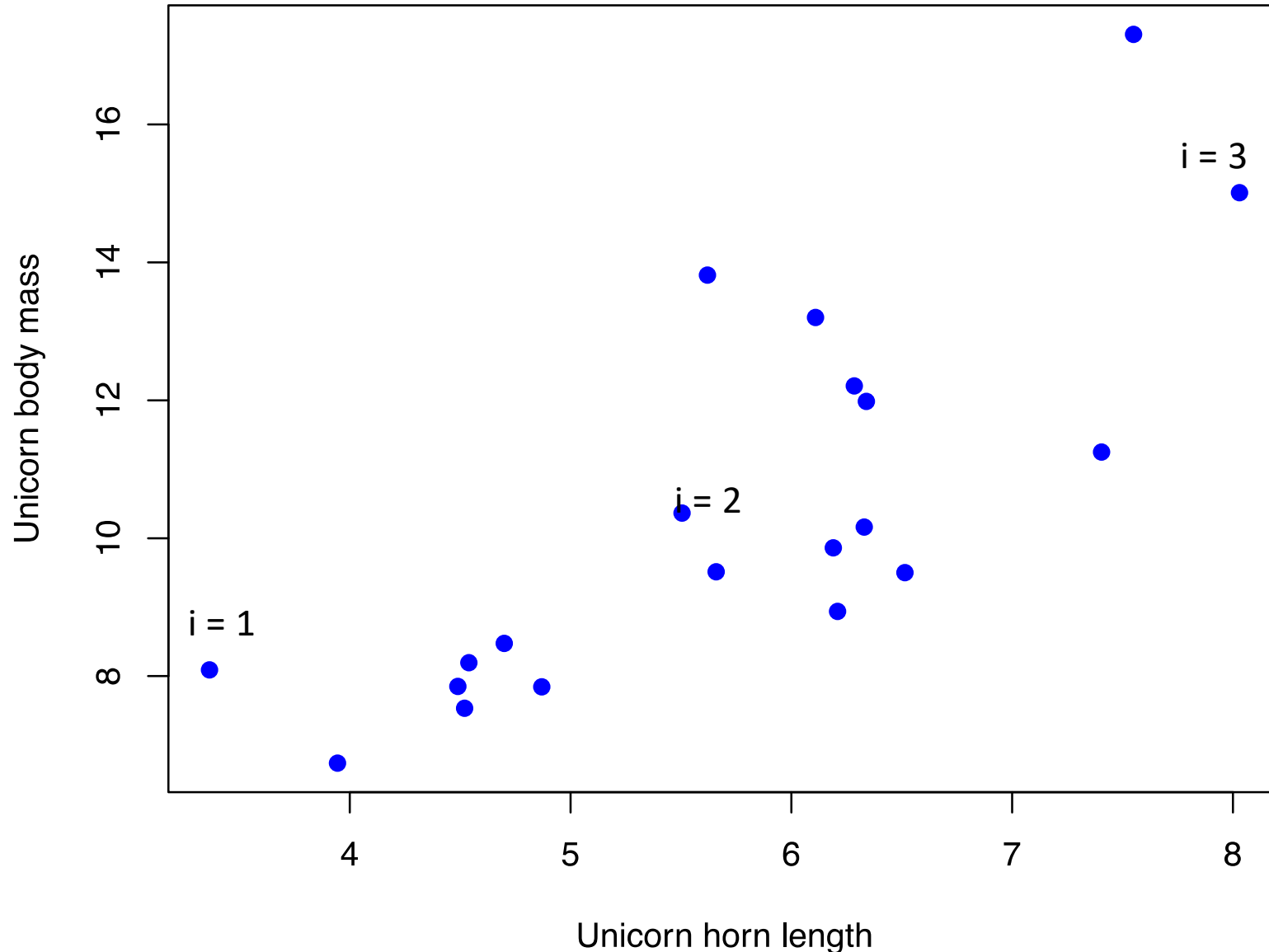
Linear models - terminology

Y = response variable, body mass

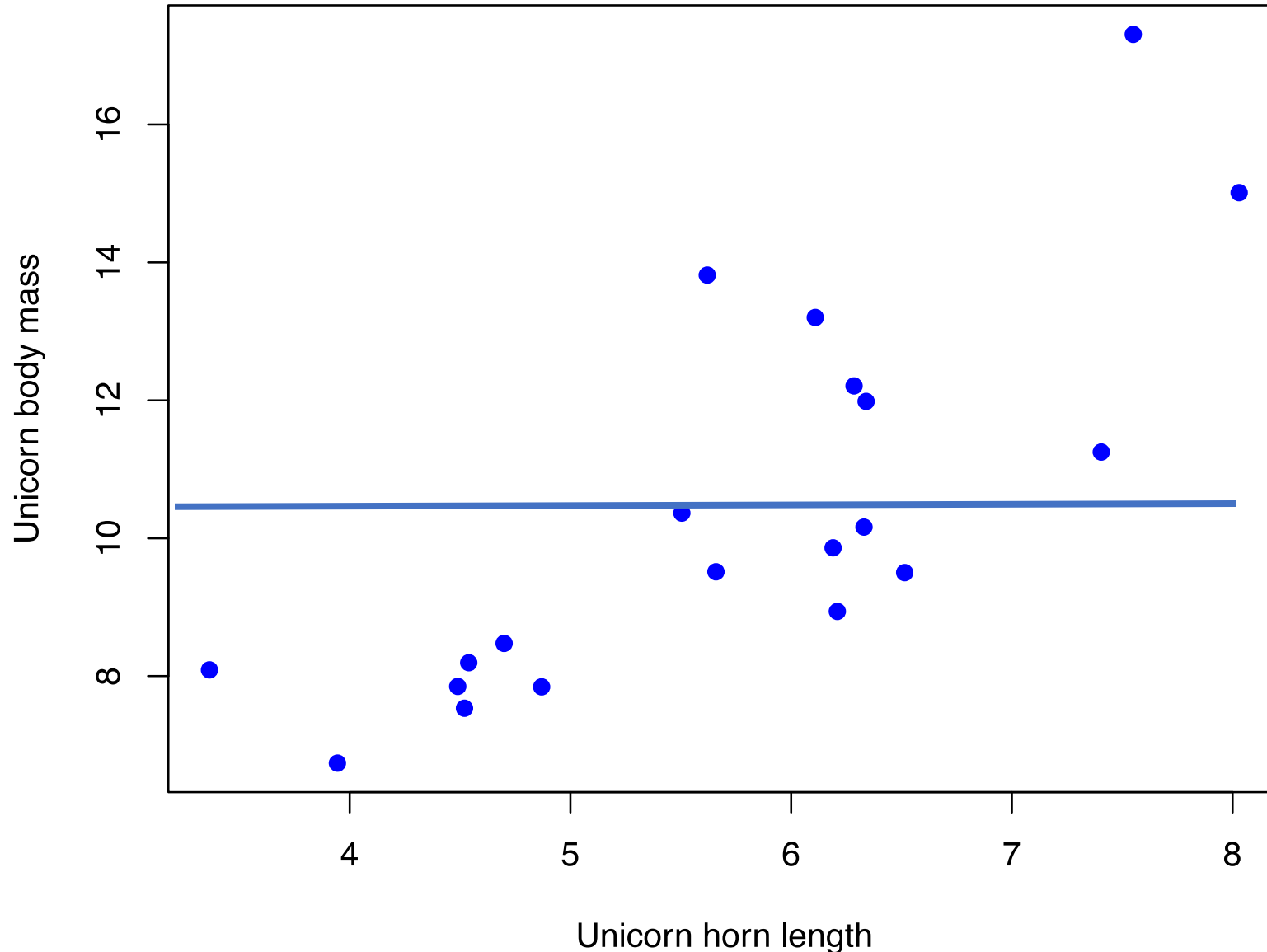
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Linear models - terminology



Y = response variable, body mass

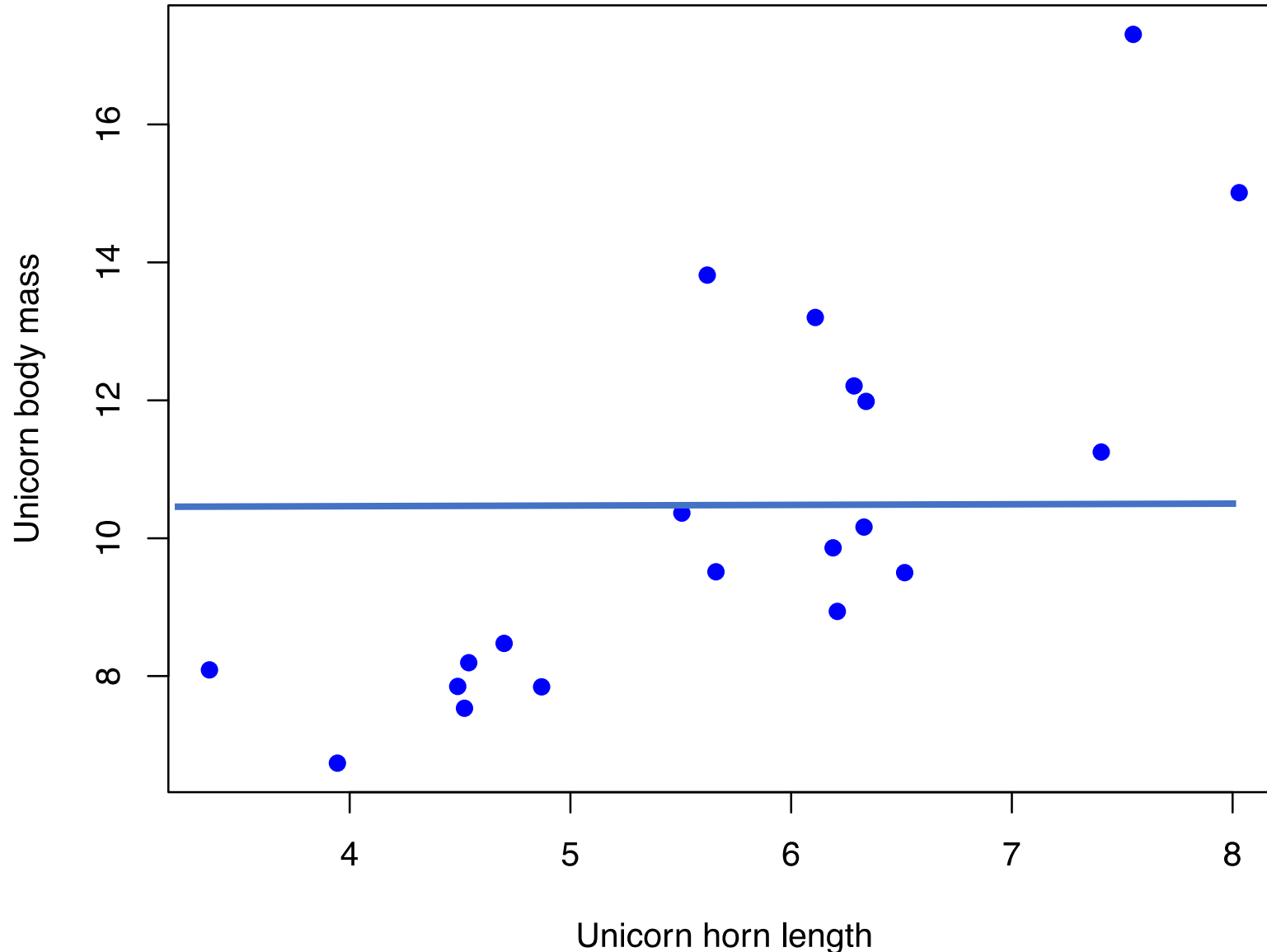
X = explanatory variable, horn length

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$\bar{Y}_i$ = EXPECTED

Linear models - terminology



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X = explanatory variable, horn length

Y_i = OBSERVATIONS of body mass

X_i = OBSERVATIONS of horn length

$\bar{Y}_i$ = EXPECTED

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```
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5  Ratline_Slinger_Rose Female 11.983053      6.340      0 4.287208 Spring    1
• 6 Betty_Striker_Boot_Rogue Female 13.199578      6.110      0 1.084253 Autumn    1
```

```
> |
```

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

$b_0 = ?$

$b_1 = ?$

ε	i
?	1
?	2
?	3
?	4
?	5

```
> head(uni)
```

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•

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$b_0 = ?$

$b_1 = ?$

ε	i
?	1
?	2
?	3
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```
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	Unicorn	Gender	Bodymass	Hornlength	Pregnant	Height	Season	Glizz	ε
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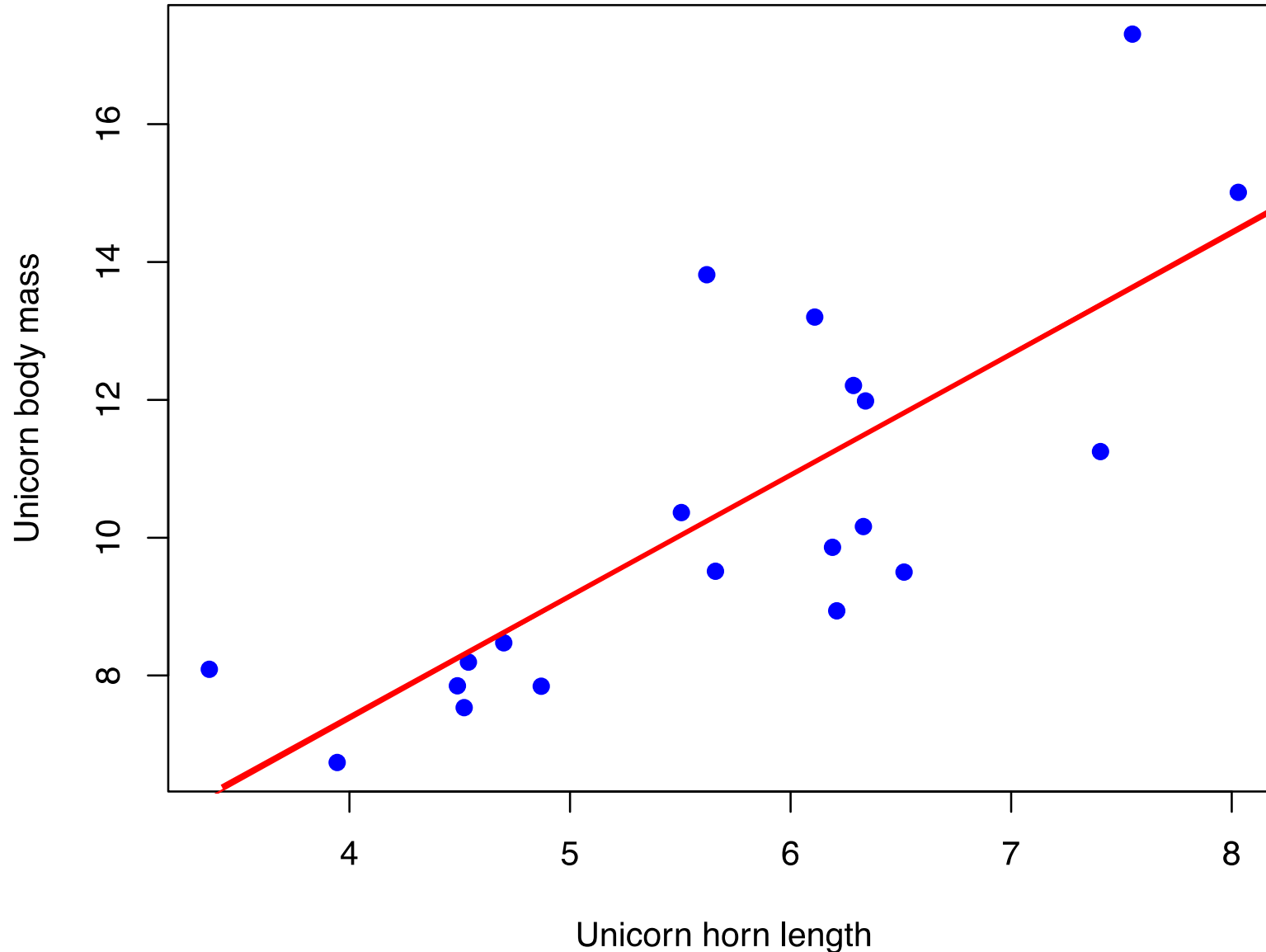
```
> |
```

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

$b_0 = ?$

$b_1 = ?$

Linear models - terminology



Y = response variable, body mass

X = explanatory variable, horn length

Y_i = OBSERVATIONS of body mass

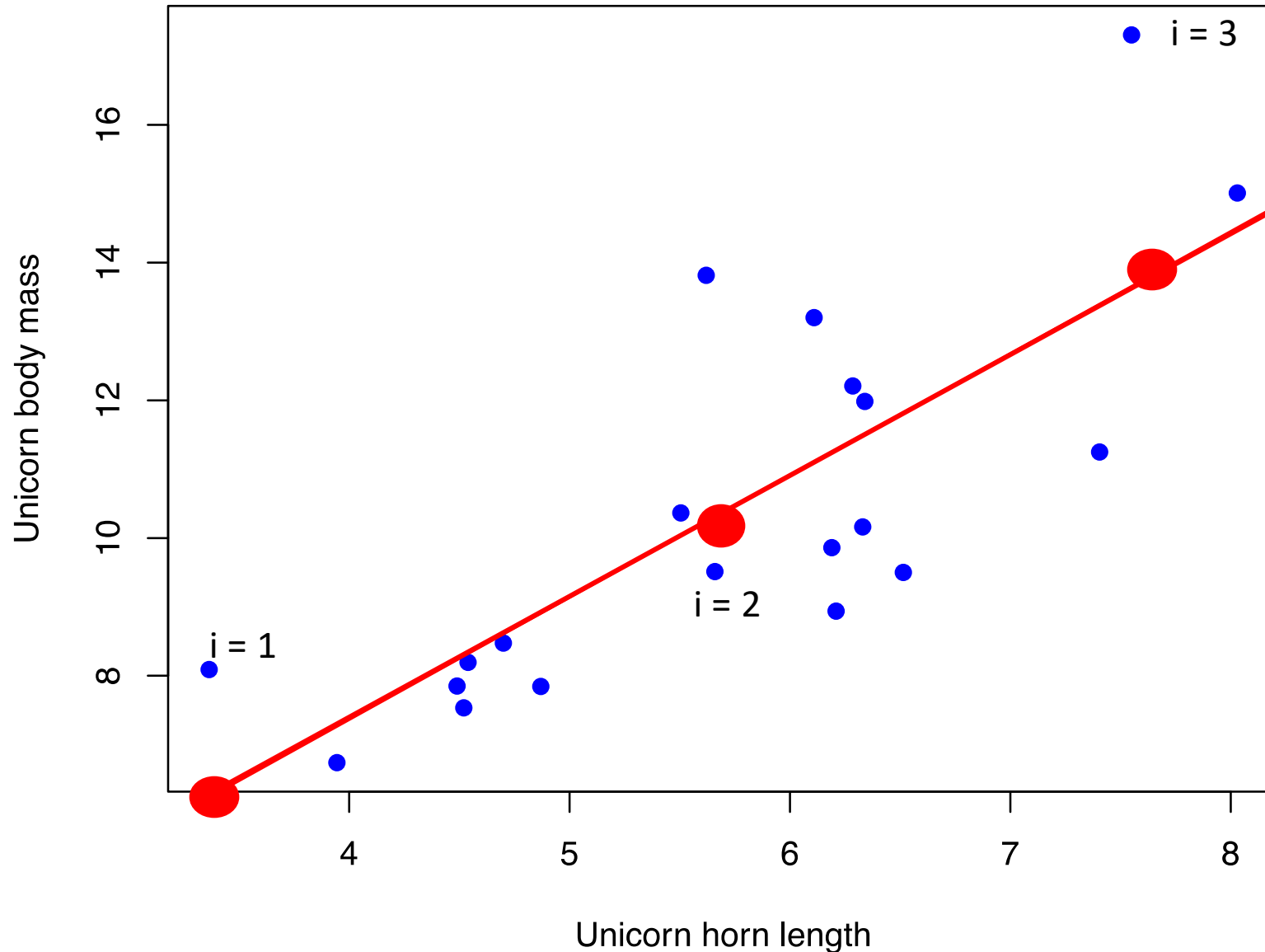
X_i = OBSERVATIONS of horn length

$\bar{Y}_i$ = EXPECTED

b_0 and b_1 , together with X_i , describe $\hat{Y}_i$

$\hat{Y}_i$ = PREDICTED

Linear models - terminology



Y = response variable, body mass

X = explanatory variable, horn length

Y_i = OBSERVATIONS of body mass

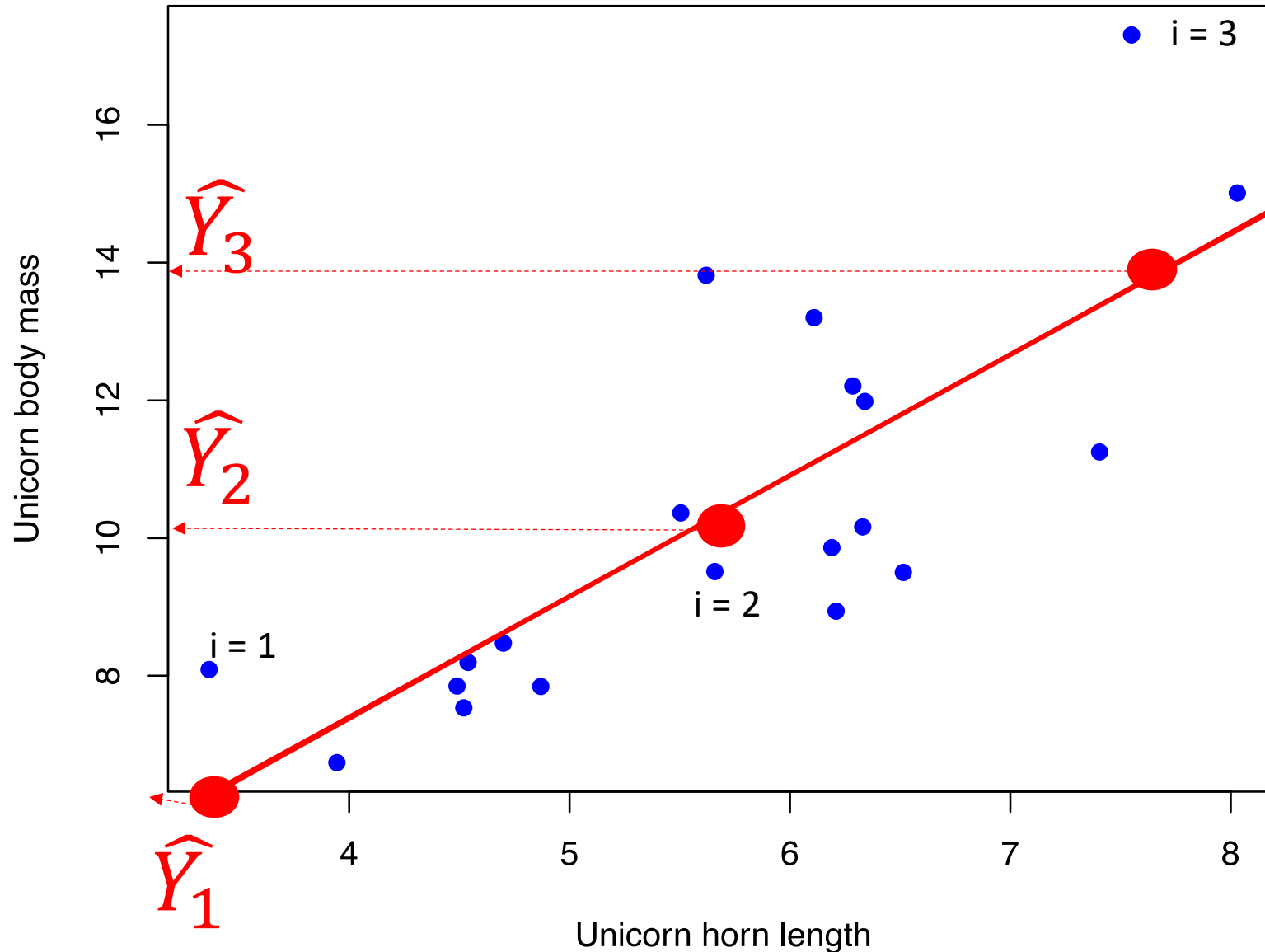
X_i = OBSERVATIONS of horn length

$\bar{Y}_i$ = EXPECTED

b_0 and b_1 , together with X_i , describe $\hat{Y}_i$

$\hat{Y}_i$ = PREDICTED

Linear models - terminology



Y = response variable, body mass

X = explanatory variable, horn length

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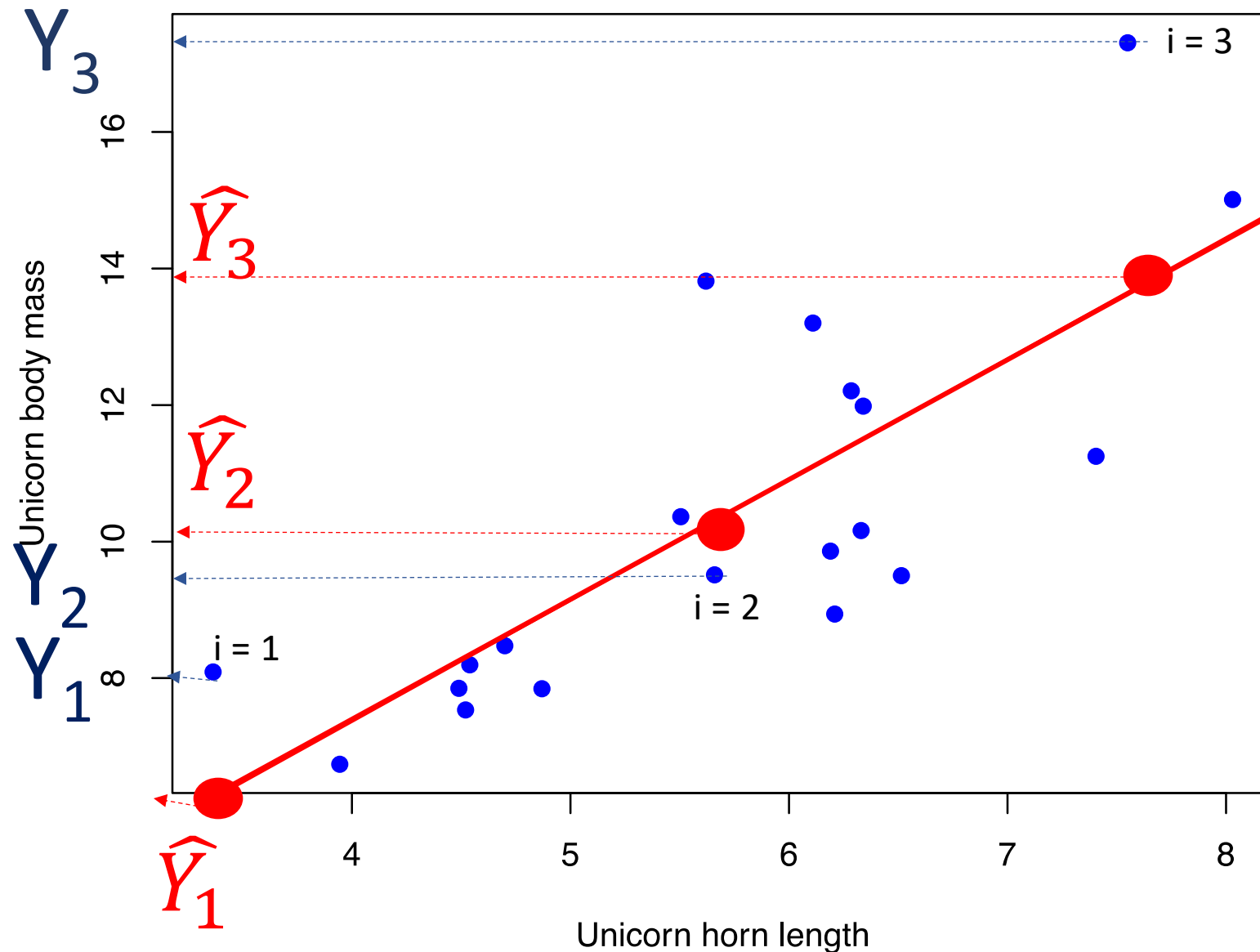
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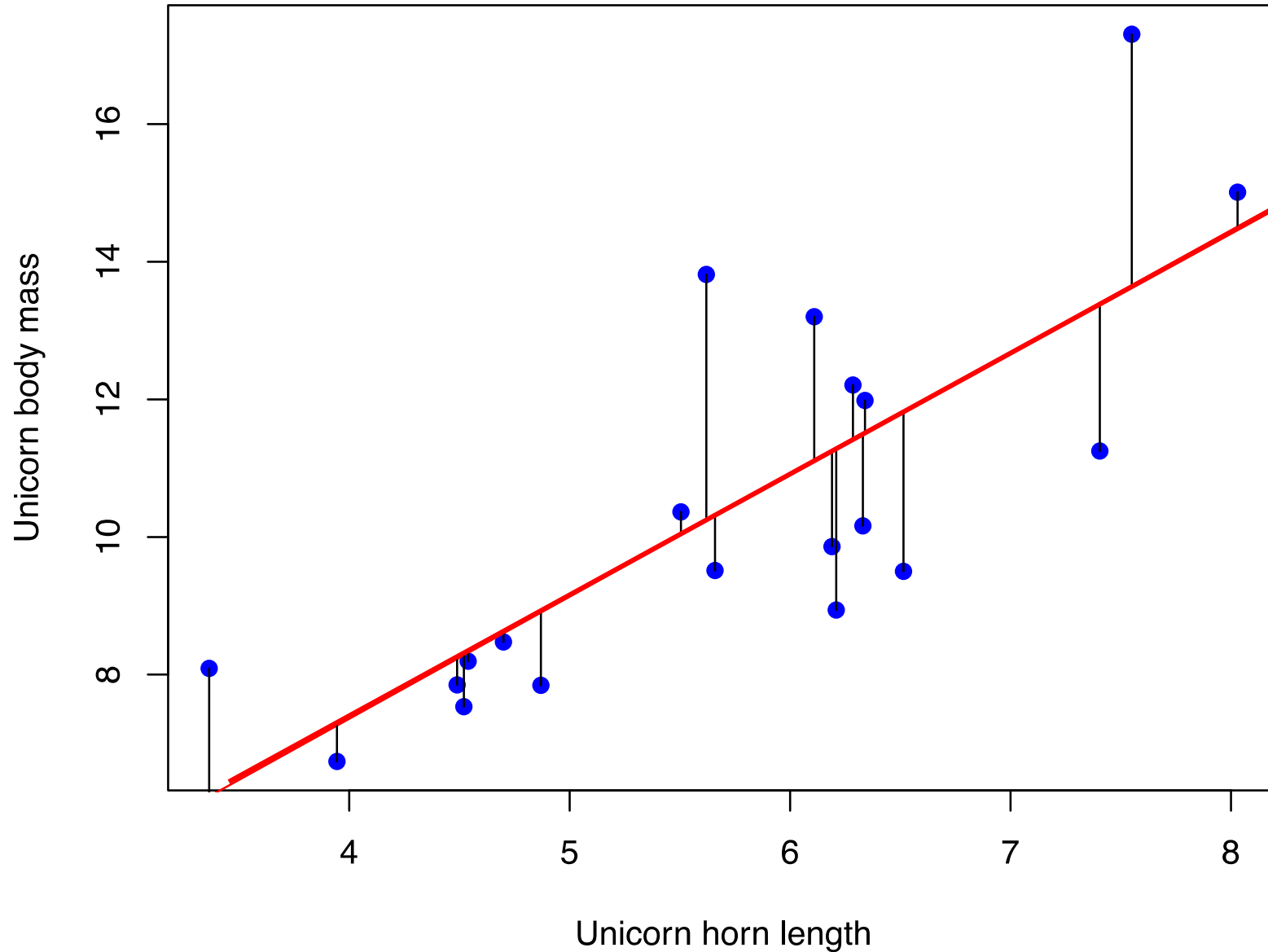
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$\hat{Y}_i$ = PREDICTED

ε_i = ERROR, residuals

Linear model predictors

- Continuous

Linear model predictors

- Continuous
- Categorical

Linear model predictors

- Continuous
- Categorical

$$y_i = b_0 + b_1 x_i + \varepsilon_i$$

y i
5 1
3 2
6 3
10 4
4 5

$b_0 = ?$

$b_1 = ?$

x	i	ε	i
blue	1	?	1
red	2	?	2
blue	3	?	3
blue	4	?	4
yellow	5	?	5

Linear model predictors

- Continuous
- Categorical
- → GENERAL linear models

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$$y_i = b_0 + b_1x_{i1} + b_2x_{i2} + b_3x_{i3} + \cdots + \varepsilon_i$$

Linear model predictors

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- Categorical
- → GENERAL linear models

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Linear model predictors

- Continuous
- Categorical
- Interactions between variables

$$y_i = b_0 + b_1x_{i1} + b_2x_{i2} + b_3x_{i1}x_{i2} + \varepsilon_i$$

Linear model predictors

- Continuous
- Categorical
- Interactions between variables
- Squared terms

$$y_i = b_0 + b_1x_{i1} + b_2x_{i1}^2 + b_3x_{i2} + b_4x_{i1}x_{i2} + \varepsilon_i$$

Linear models – advice for reporting

- **Report:**
 - **model structure**
 - **Parameter estimates + precision**
 - **Effect sizes**
- **Less important (but still reported):**
 - P-values
 - T-values
- **Stay away from:**
 - ANOVA, ANCOVA, MANOVA – use linear models instead as they are more biologically informative and easier to interpret
 - **Reporting statistics (F, R, t, ect) WITHOUT parameter estimates**

Process

1. Outliers?

Process

1. Outliers?
2. Homogeneity of variances?

Process

1. Outliers?
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3. Normally distributed residuals?

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Process – common problems

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What to include? **HYPOTHESIS**

- Does the horn of the unicorns help them to eat more so they are heavier?

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Horn length ~ Body mass

$$Horn\ length_i = \beta_0 + \beta_1 Hornlength_i + \varepsilon_i$$

What to include? **HYPOTHESIS**

- Maybe heavier unicorns are able to grow larger horns?
 - Age?
 - Sex?
 - Size?
 - Nutrients in diet?
 - Ecology (shorter horns in denser forests, longer horns on grasslands)
 - Predator presence

Process – common problems

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How to decide which model is good?



?



Different ways to compare models

- Information criterion (AIC, BIC, DIC)
- Log-likelihood

Model selection

- Common sense!
- Information criterion (AIC, BIC, DIC)
- Likelihood ratio test
- Step wise deletion



Model selection

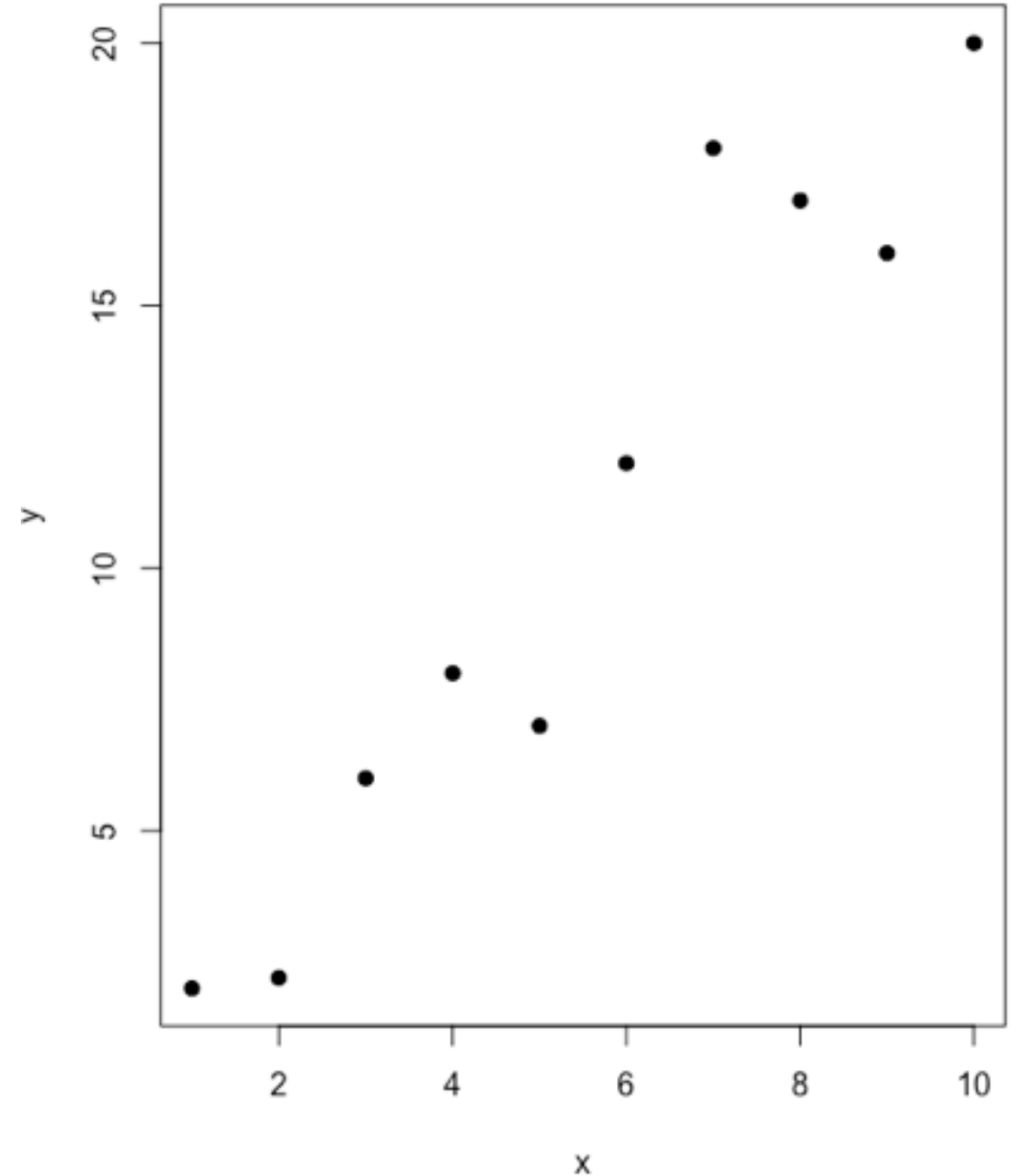
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Model selection

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- Only gives information about goodness of fit, not biology!

Model selection

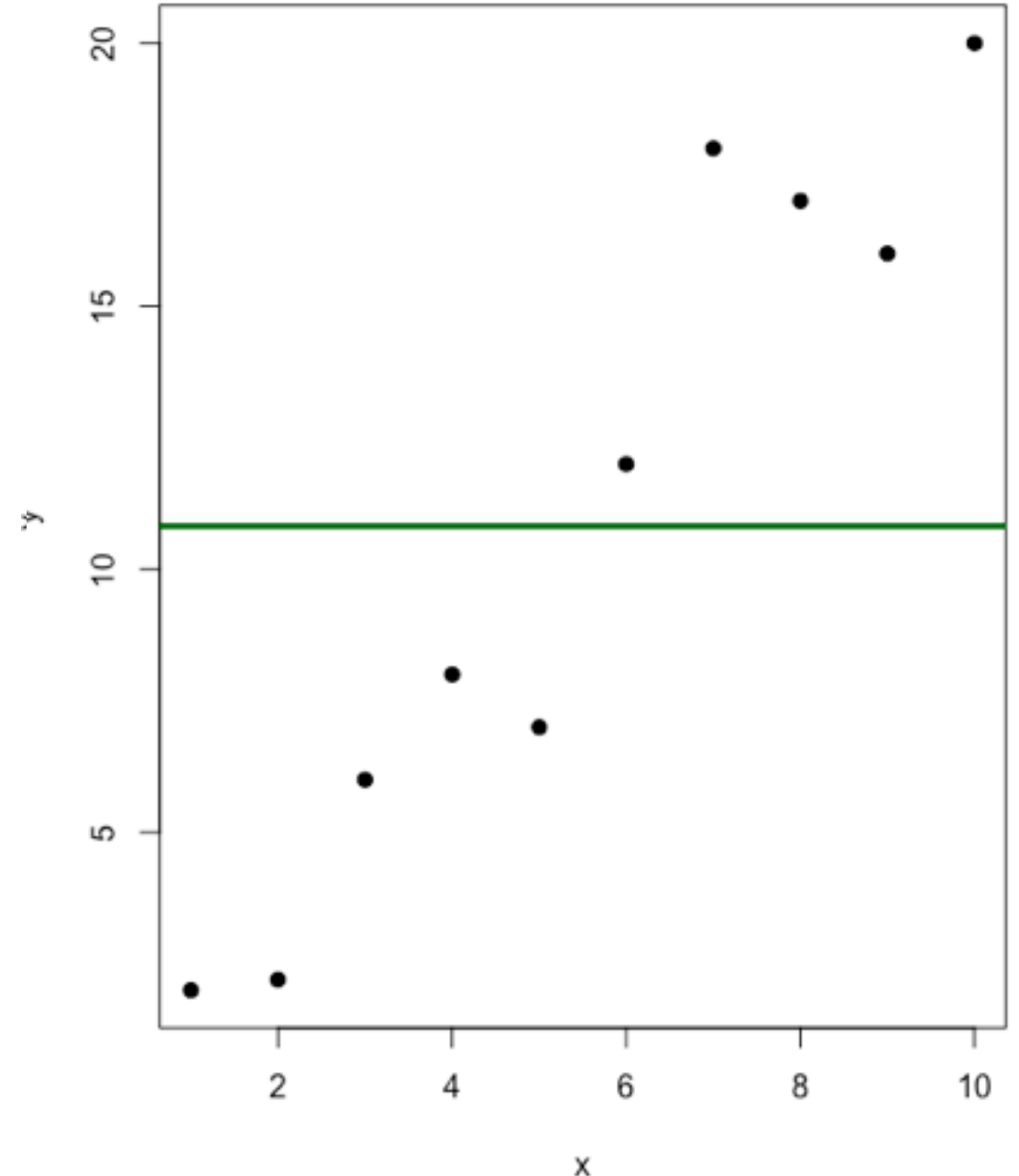
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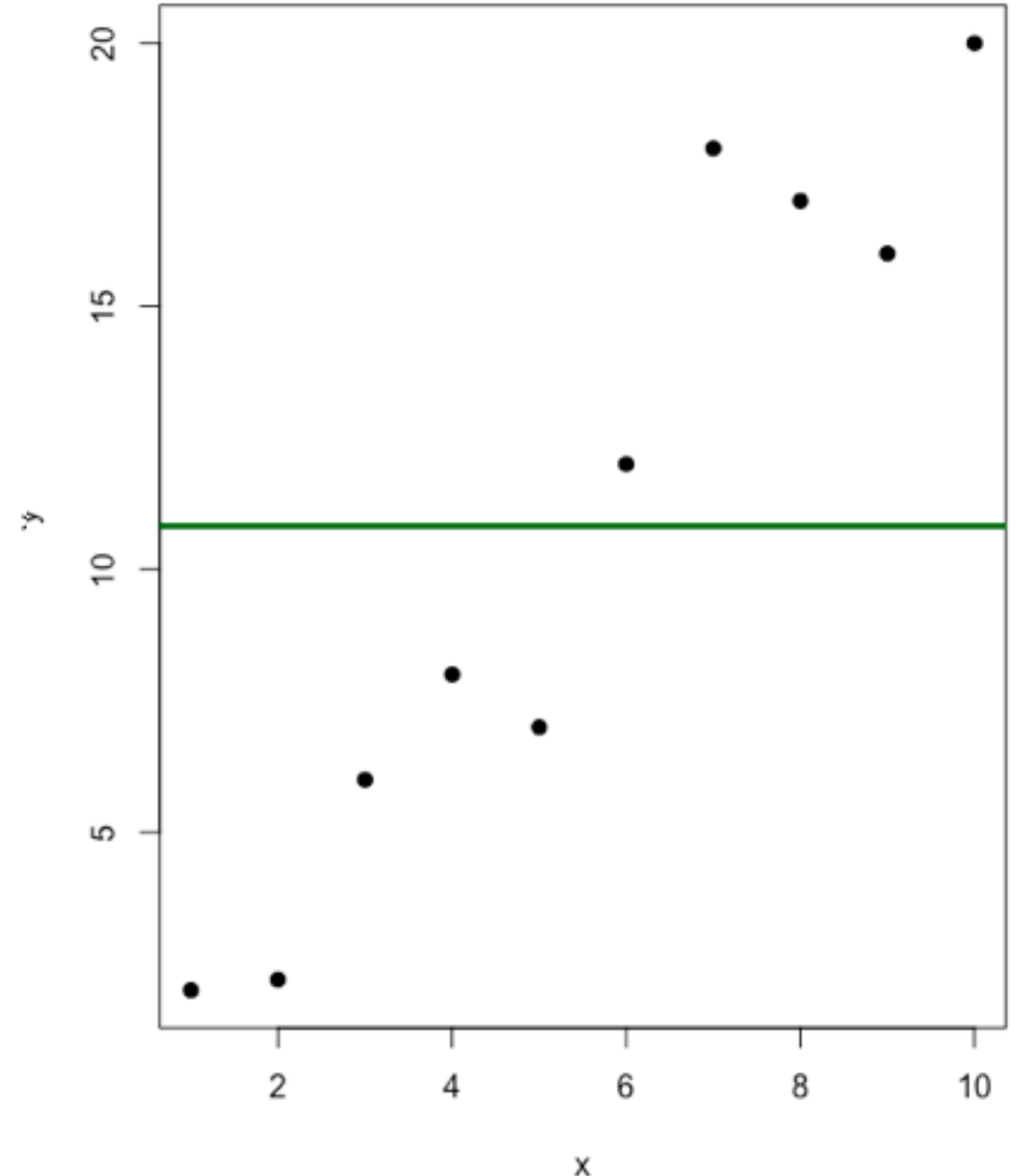
$$y_i = \beta_0$$



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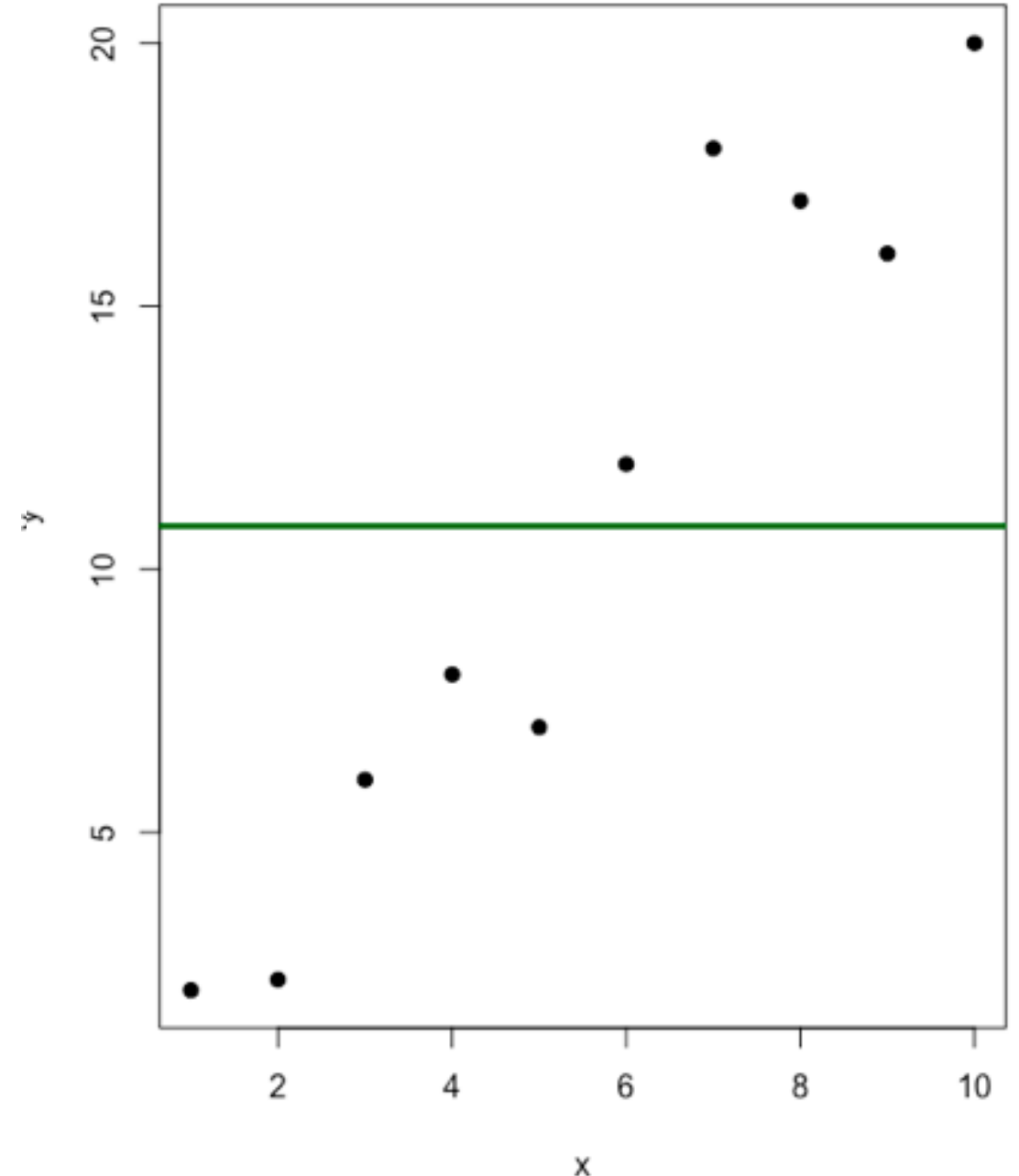
$$y_i = \beta_0, \text{ } df = n-1$$



Model selection

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$$y_i = \beta_0, \text{ df} = n - 1 = 10$$

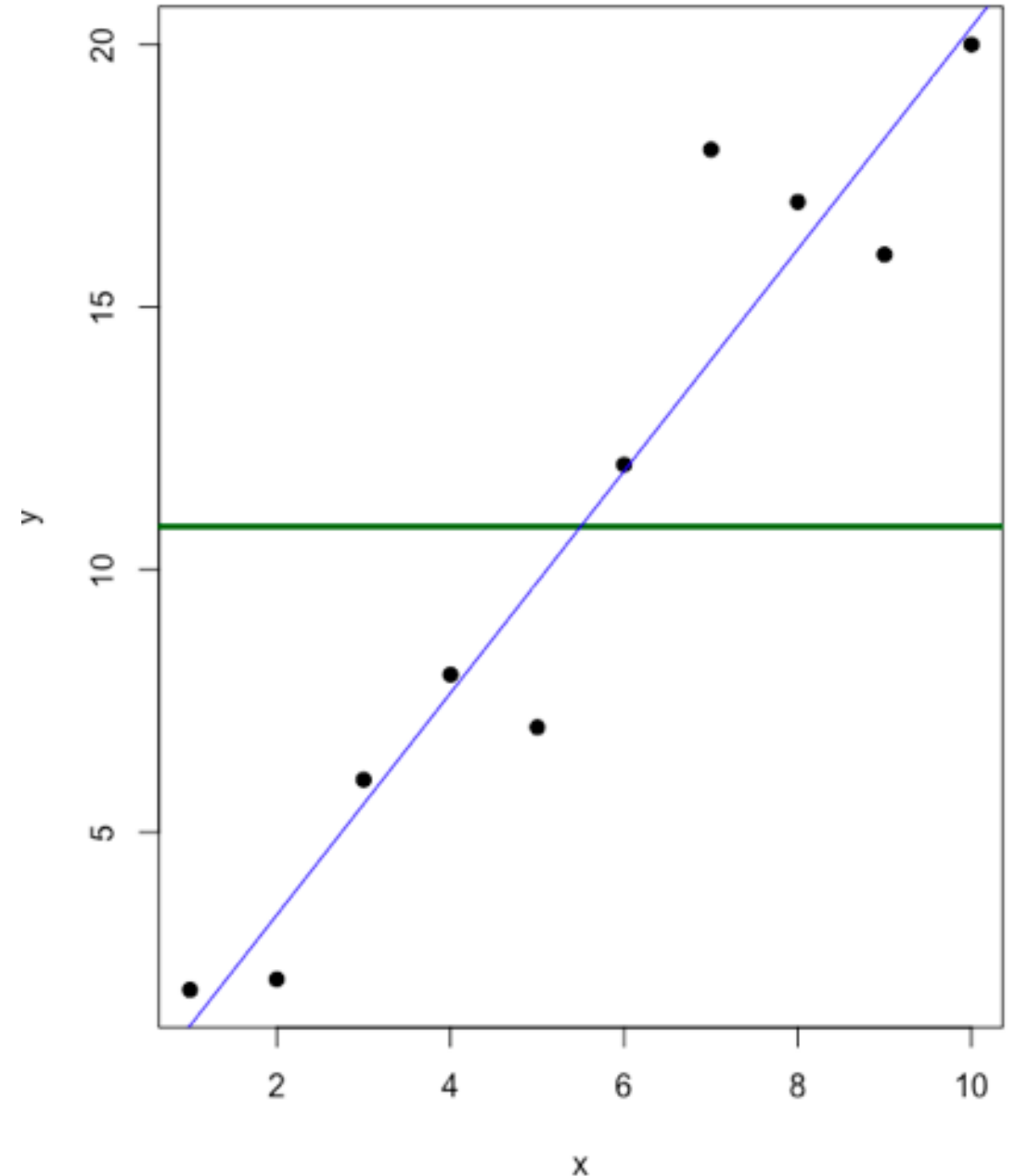


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$$y_i = \beta_0, \text{ } df = n - 1 = 10$$

$$y_i = \beta_0 + \beta_1 x_i$$

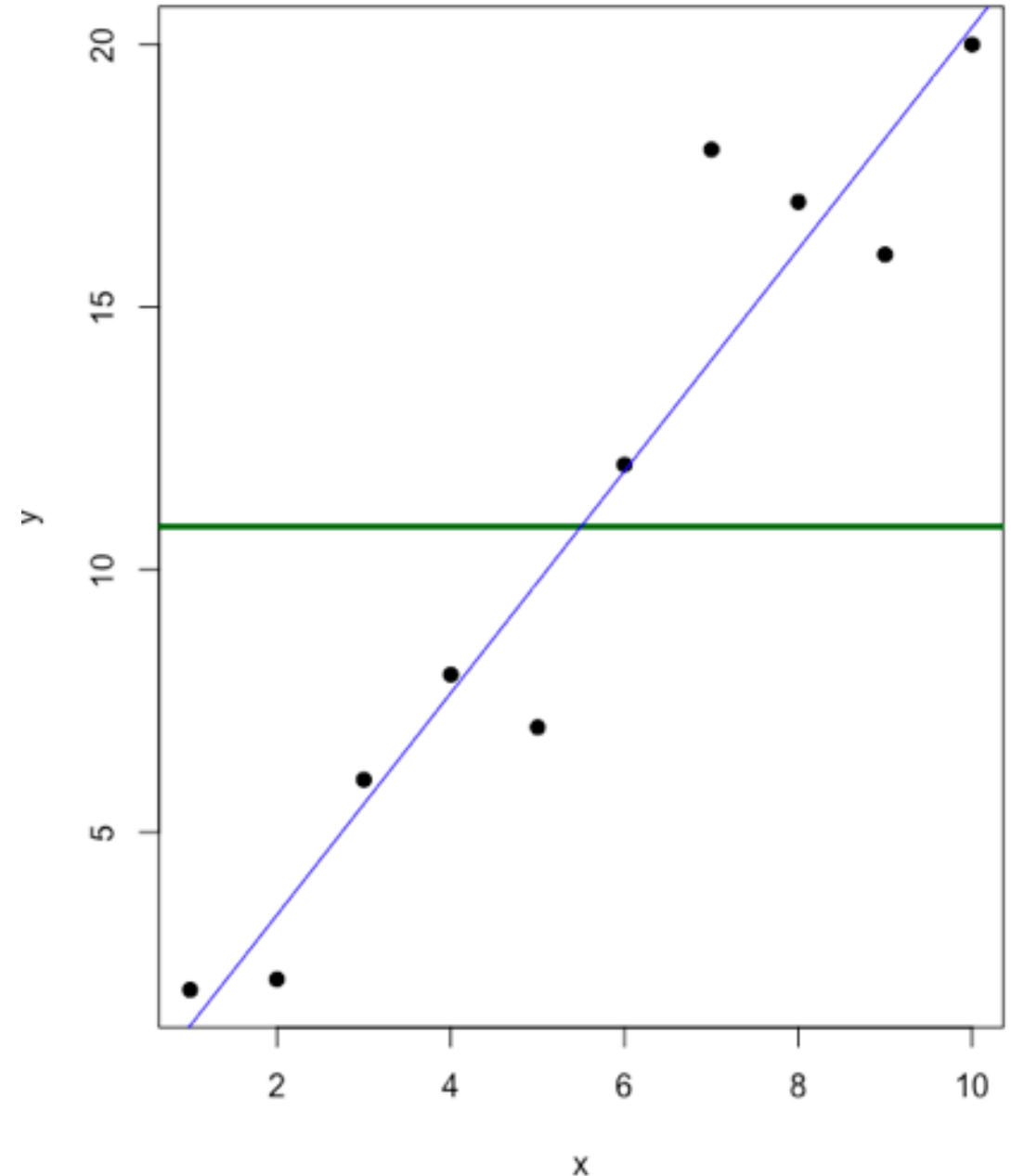


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$$y_i = \beta_0, \text{ df} = n-1 = 9$$

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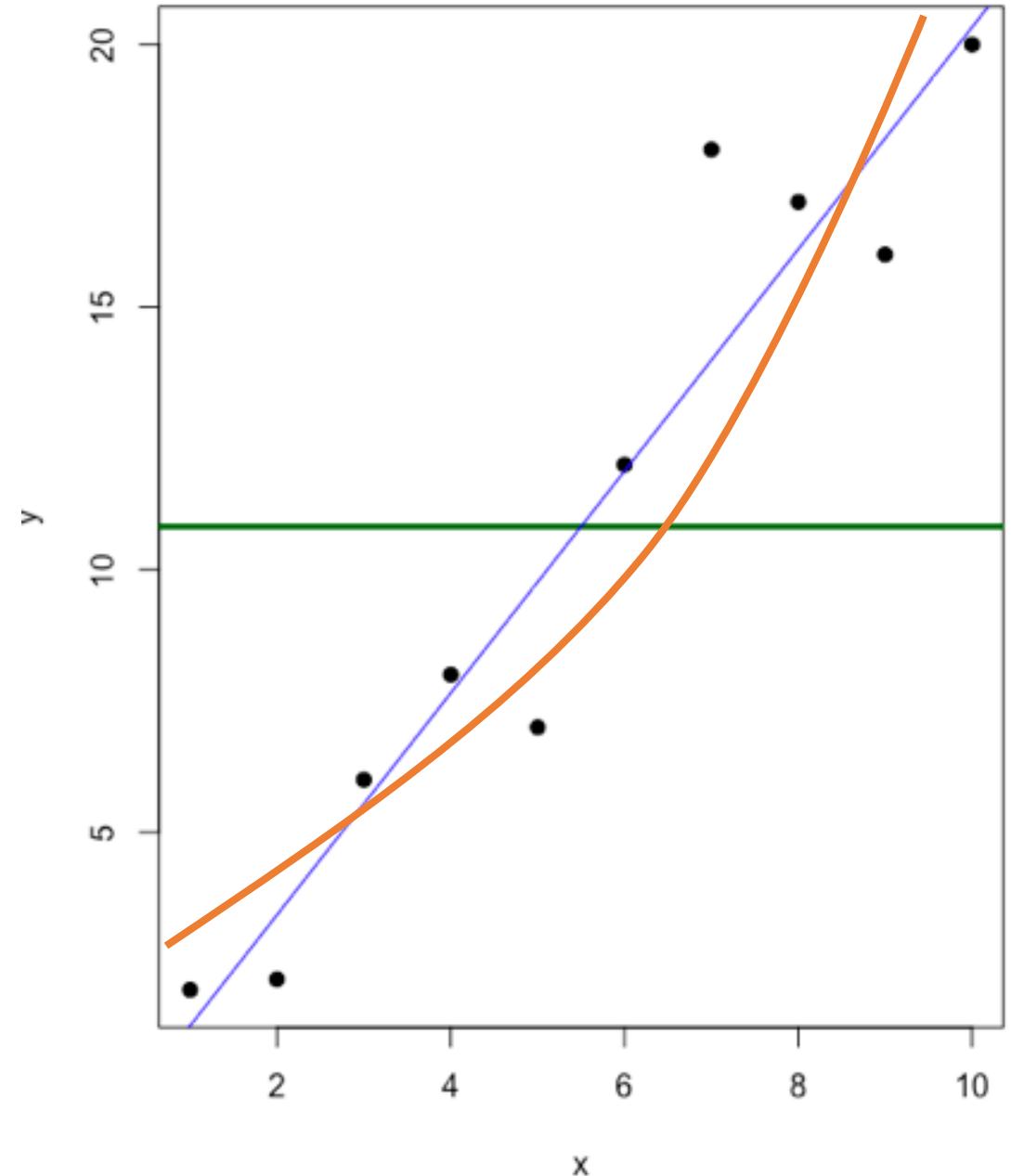
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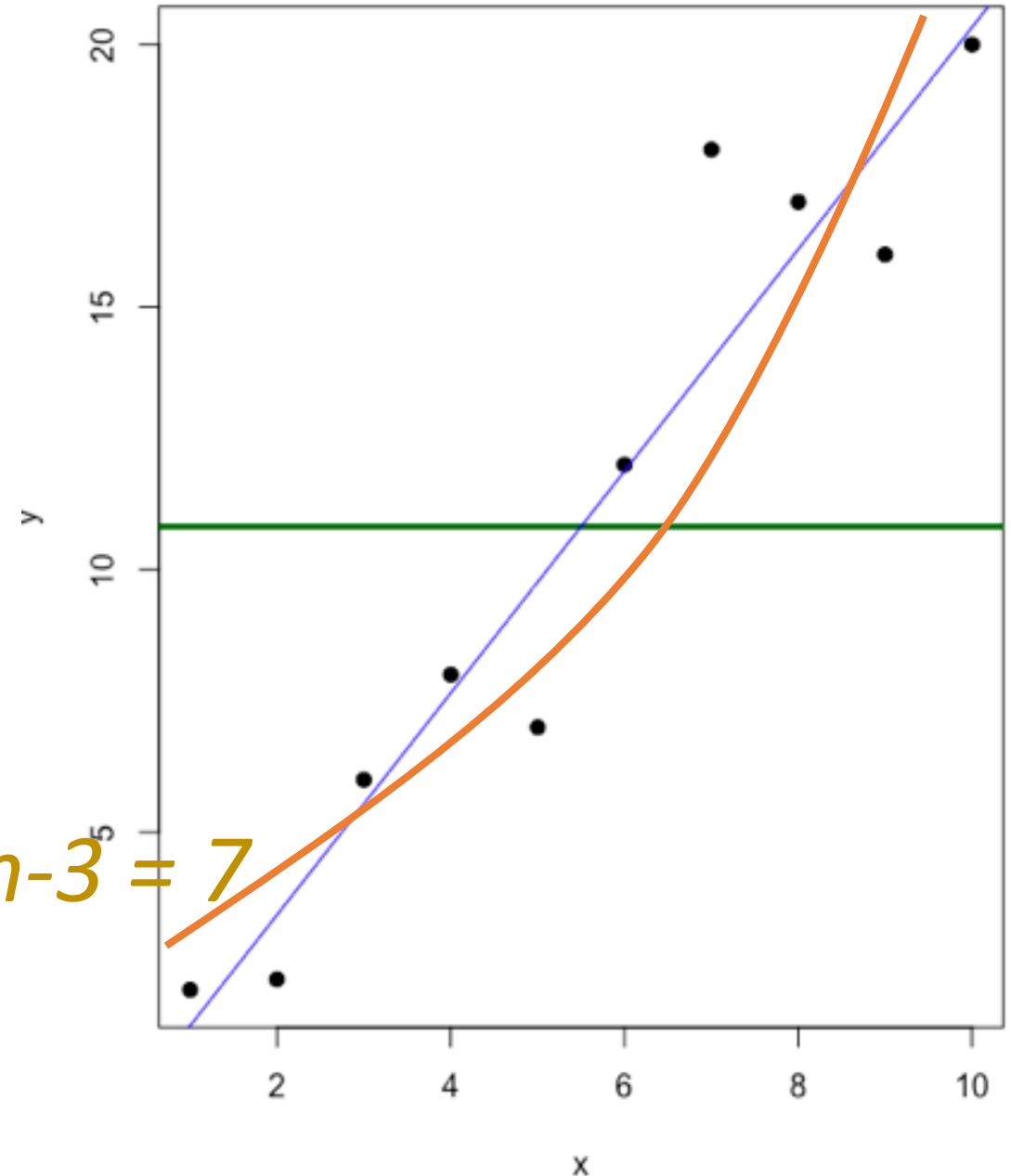
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Model selection

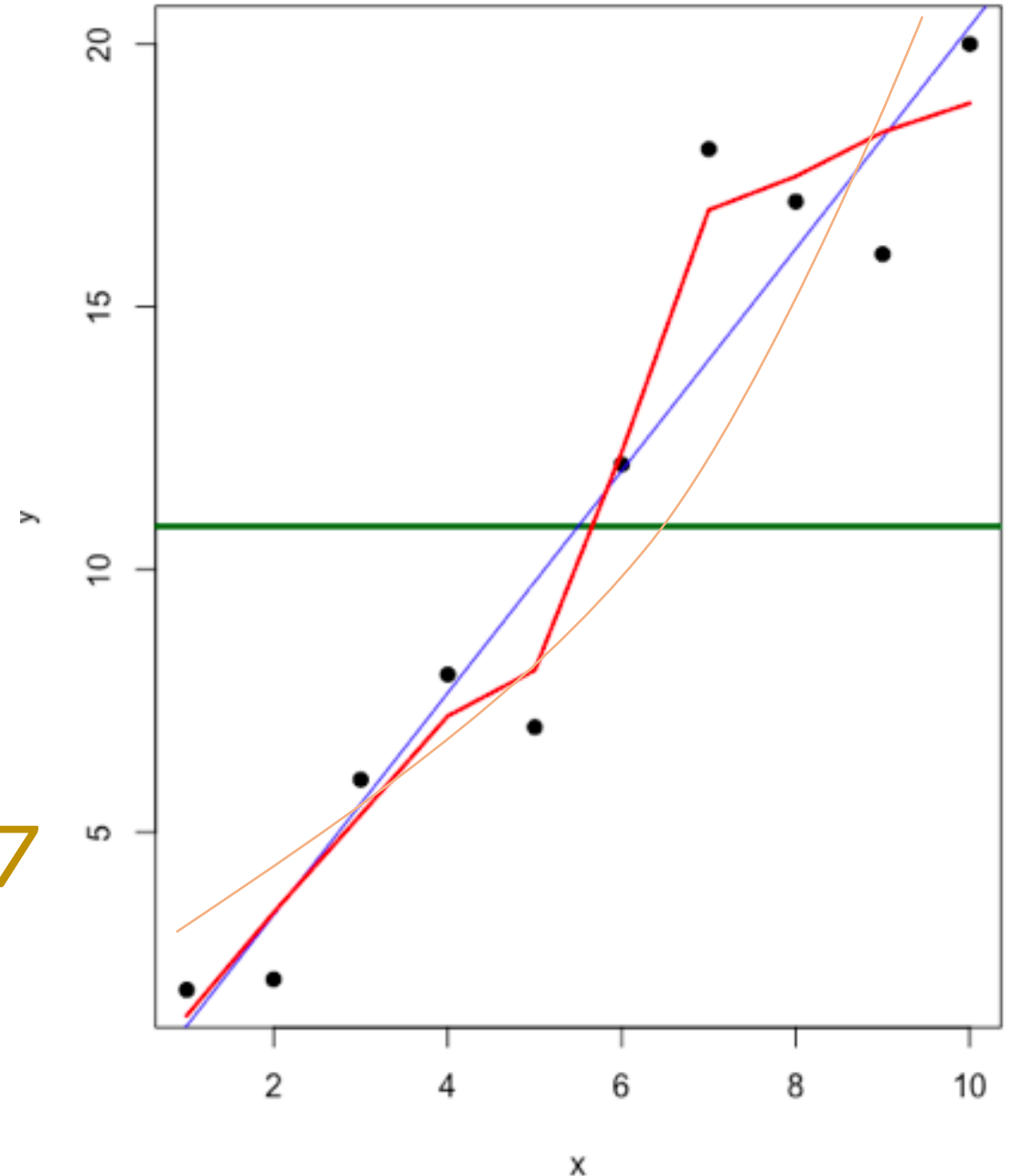
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$$y_i = \text{LOESS } f, \text{ df} = n-5 = 5$$



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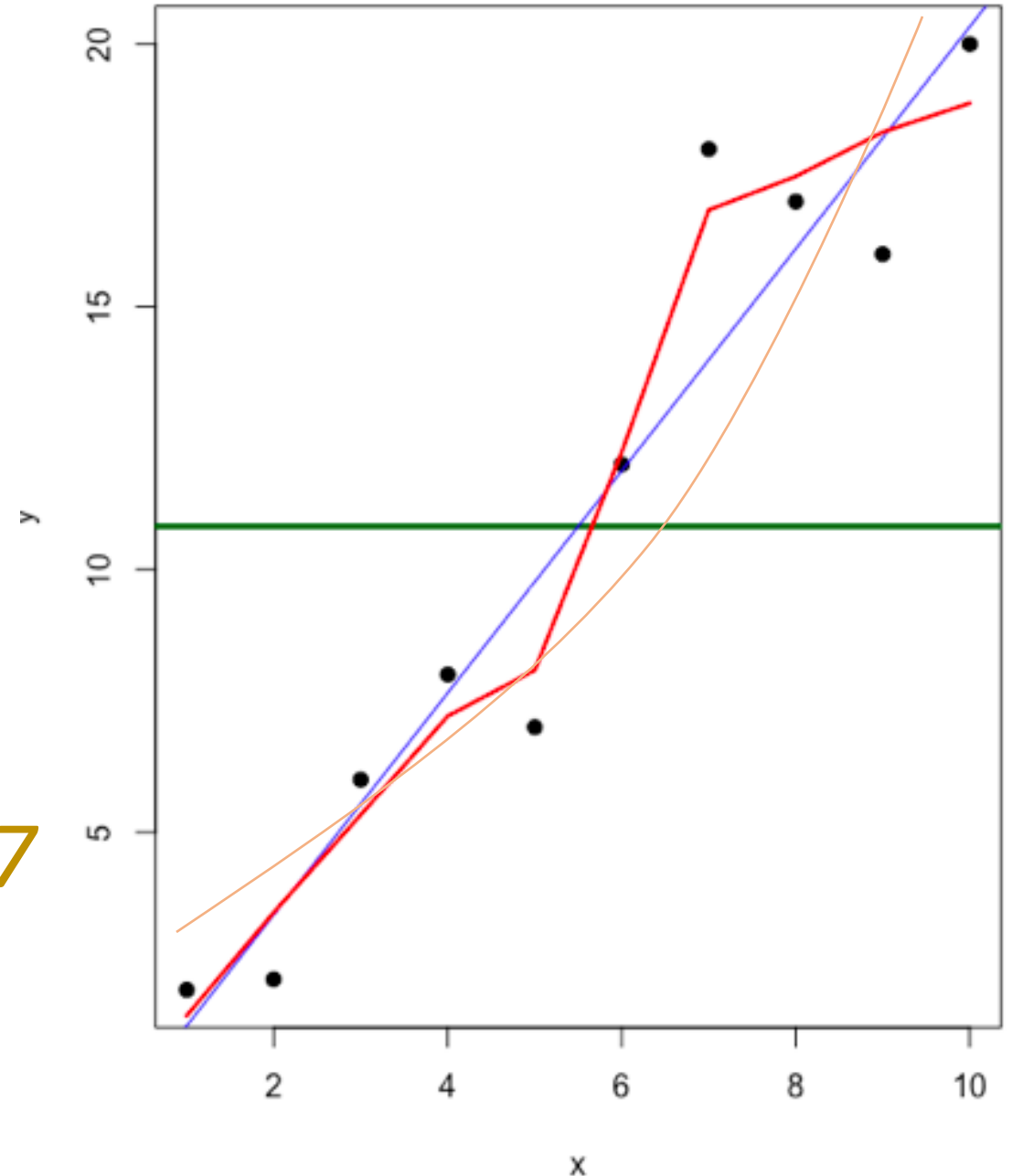
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$$y_i = y_i, \text{ df} = n-10 = 0$$



Model selection

Top – down: increasing fit

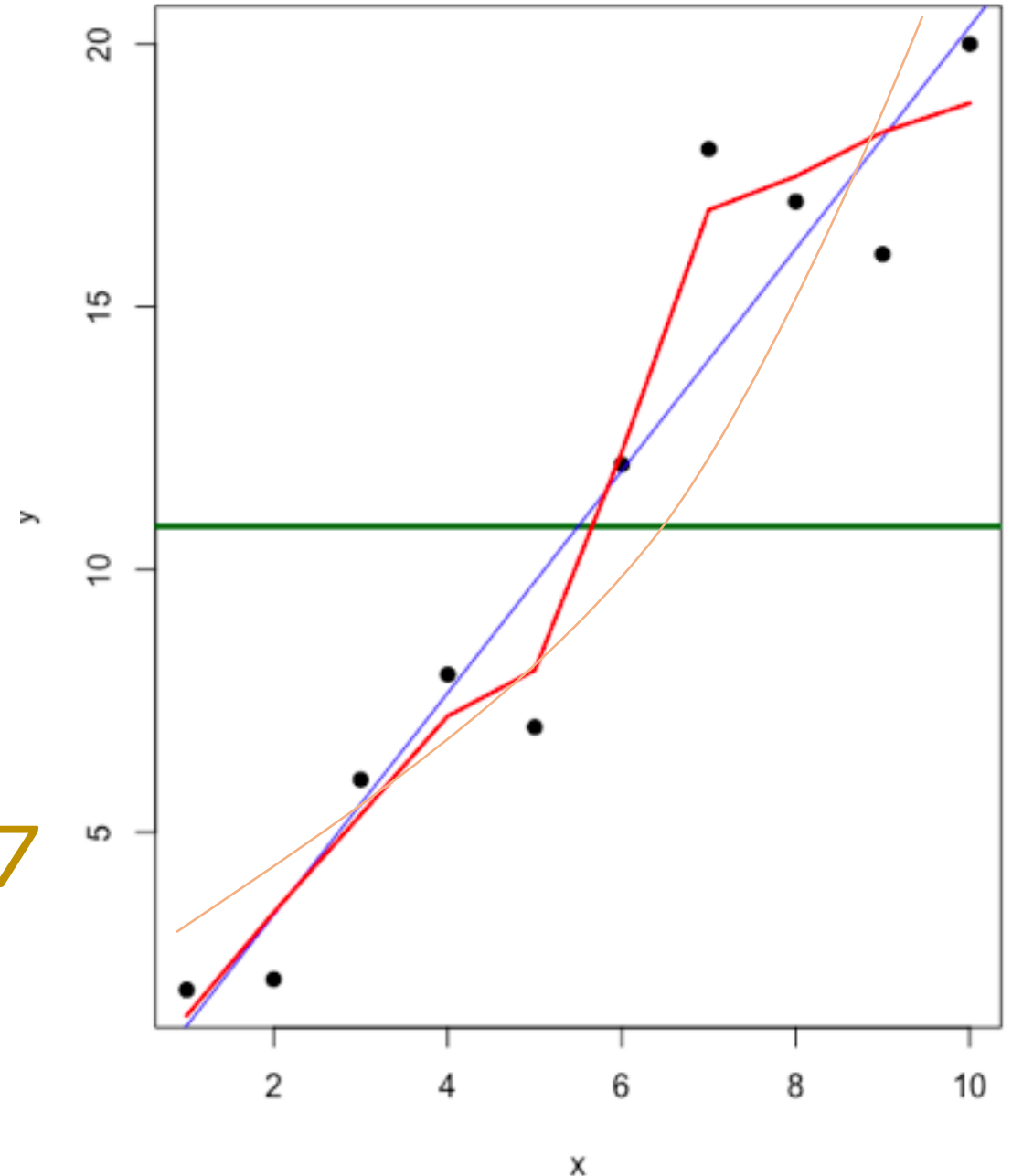
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$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2, df = 7$

$y_i = LOESS\ f, df = n-5 = 5$

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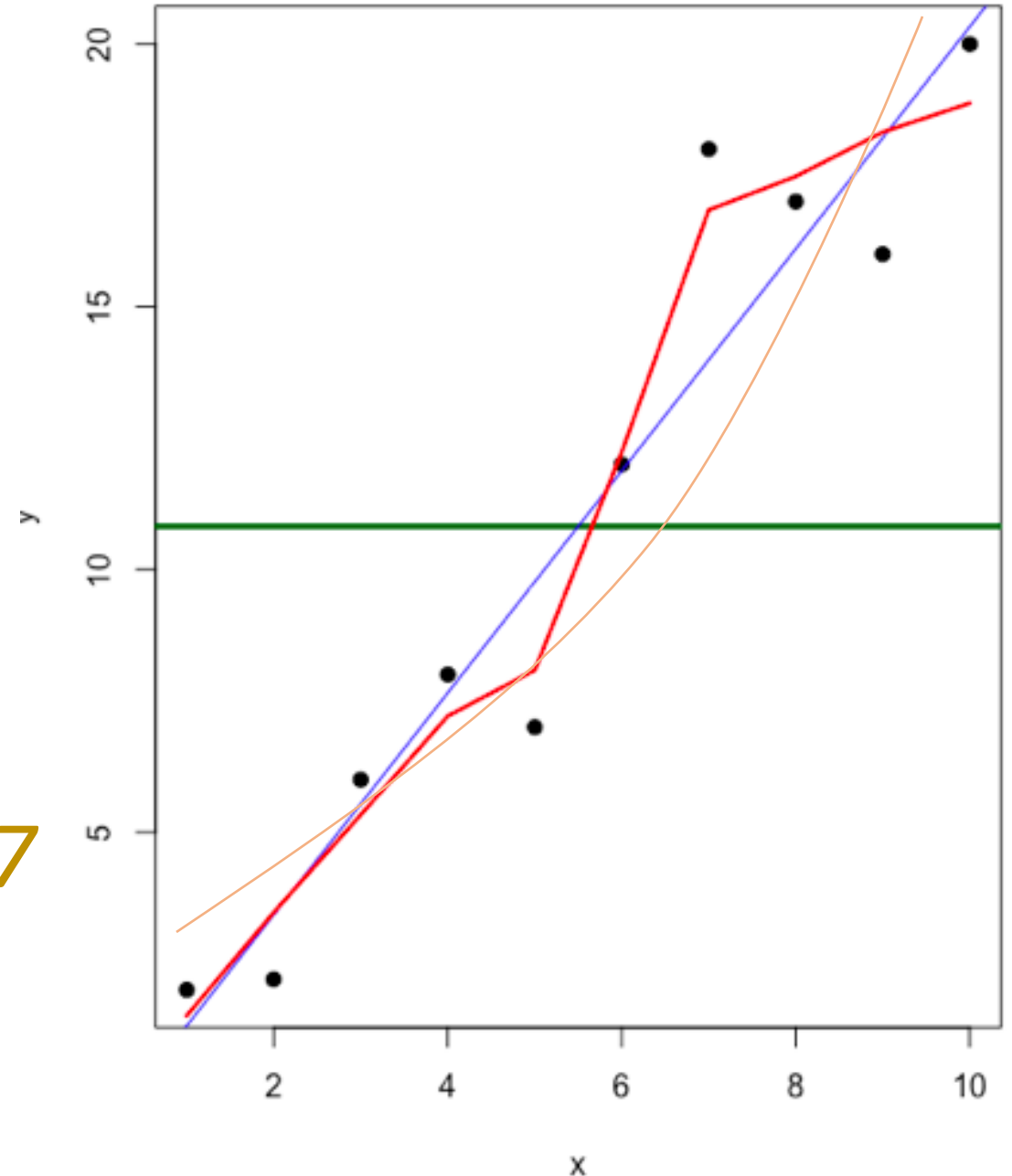


Model selection

Top – down: increasing fit

Bottom – up: loss of dfs

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↓ $y_i = y_i, df = n-10 = 0$



Model selection

Top – down: increasing fit

Bottom – up: loss of *dfs*

Trade-off explanatory power with complexity

↑

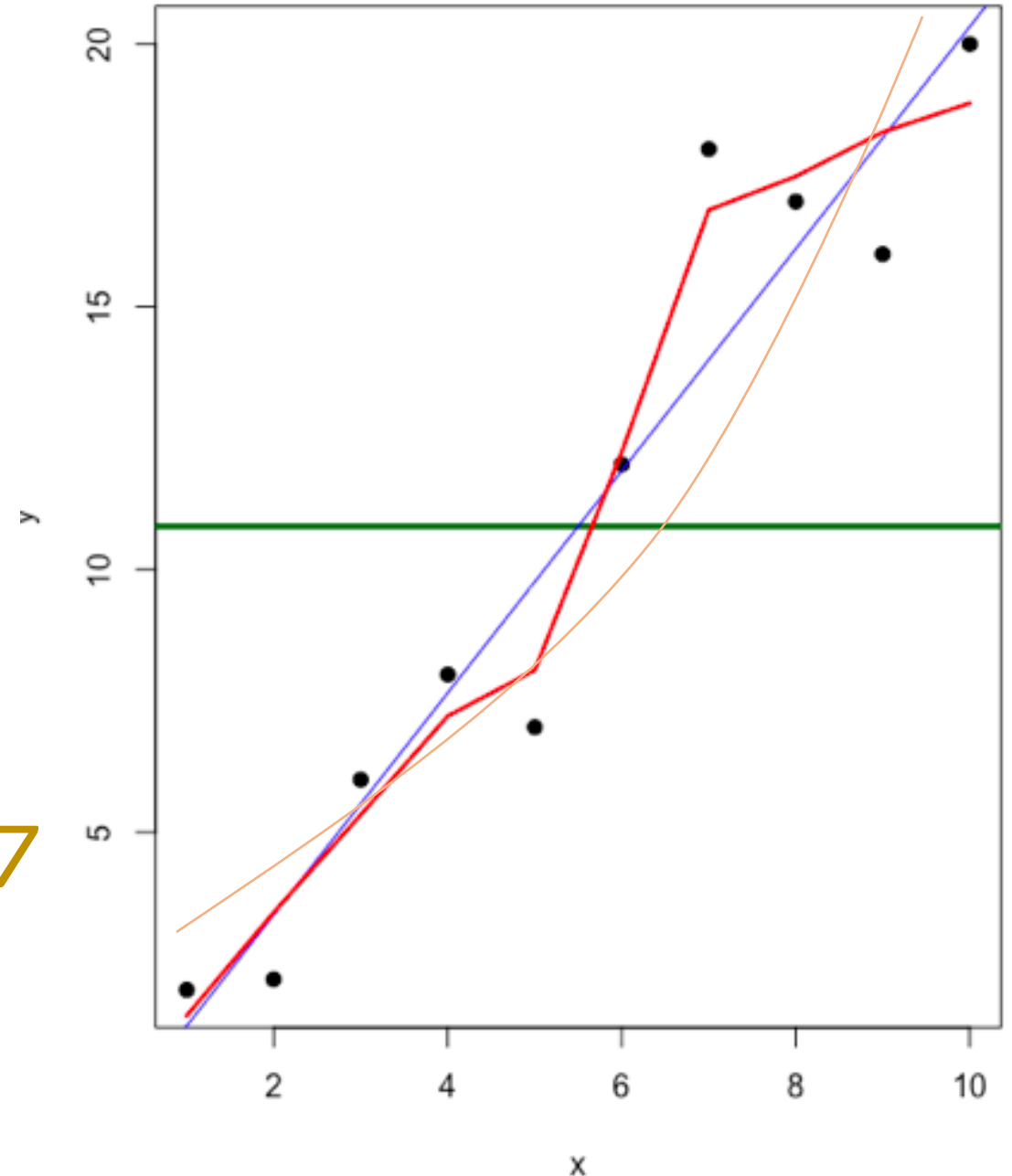
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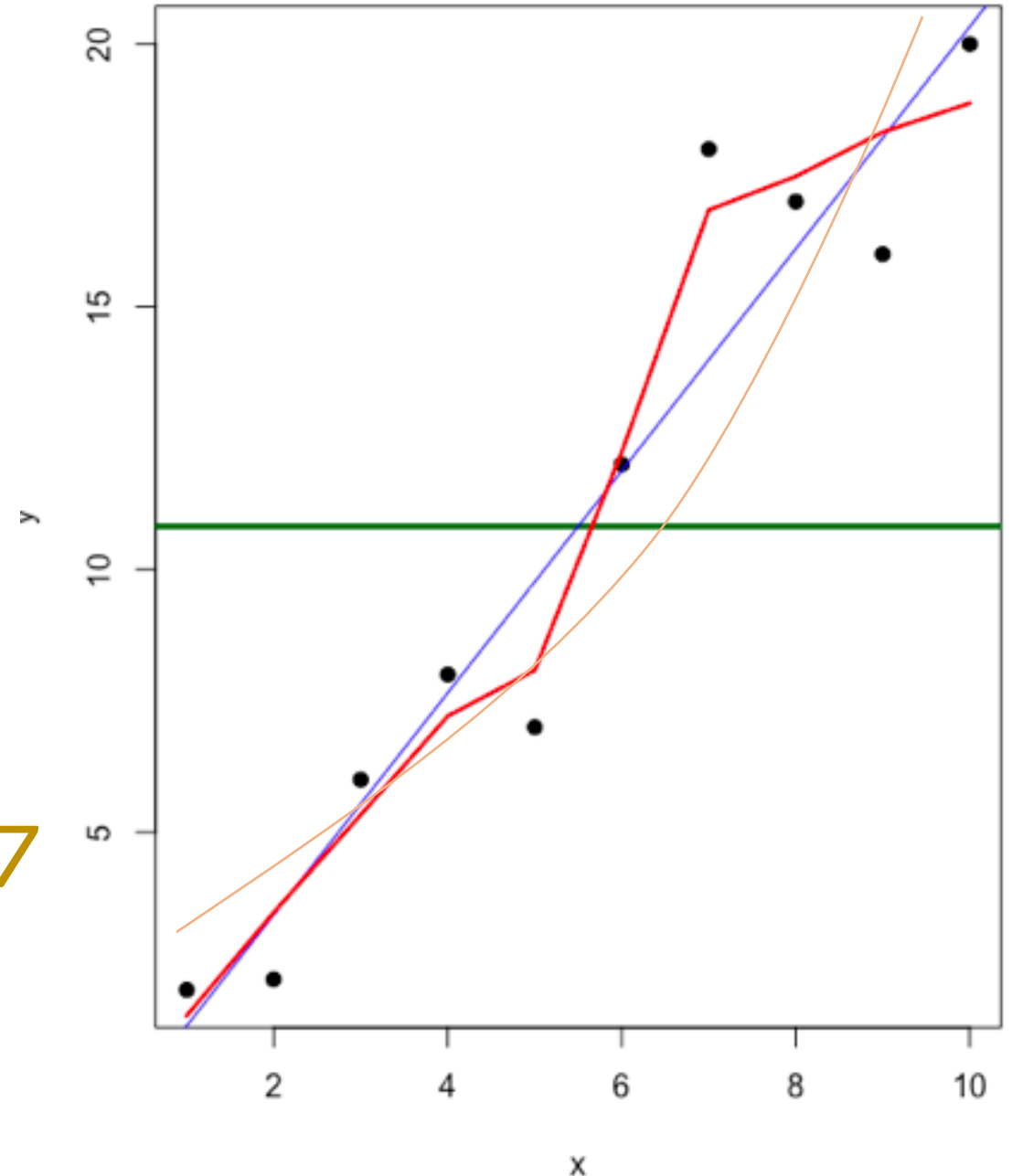
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Model selection

Which model to select?

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Model selection

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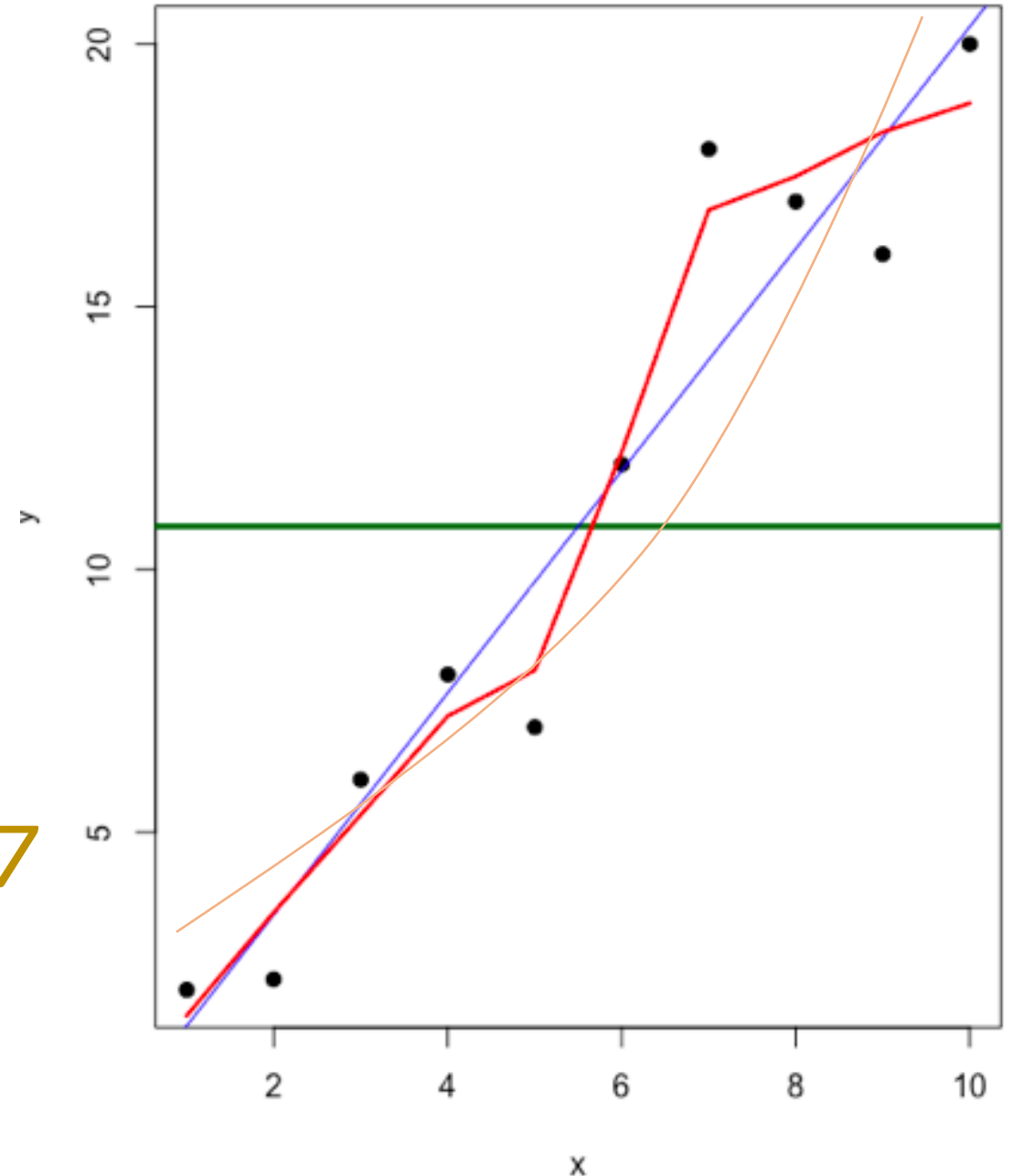
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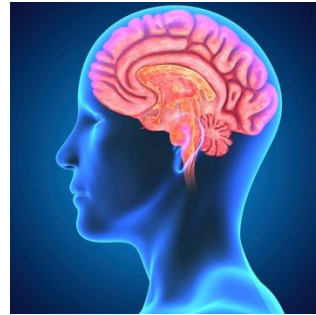
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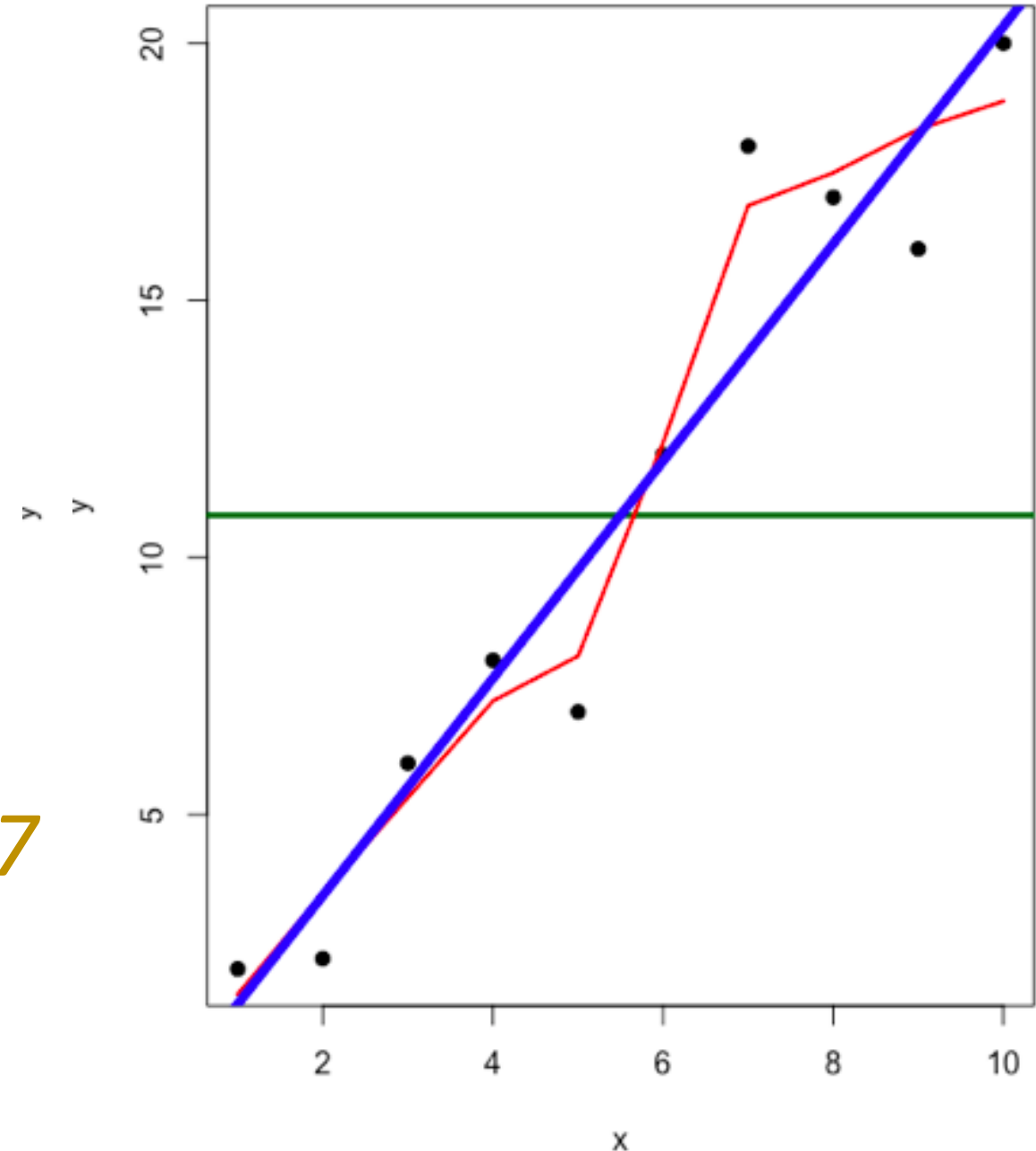


Model selection

Which model to select?



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Model selection

Which model to select?

- Sparrow parental care in relationship to offspring number



Model selection

Which model to select?

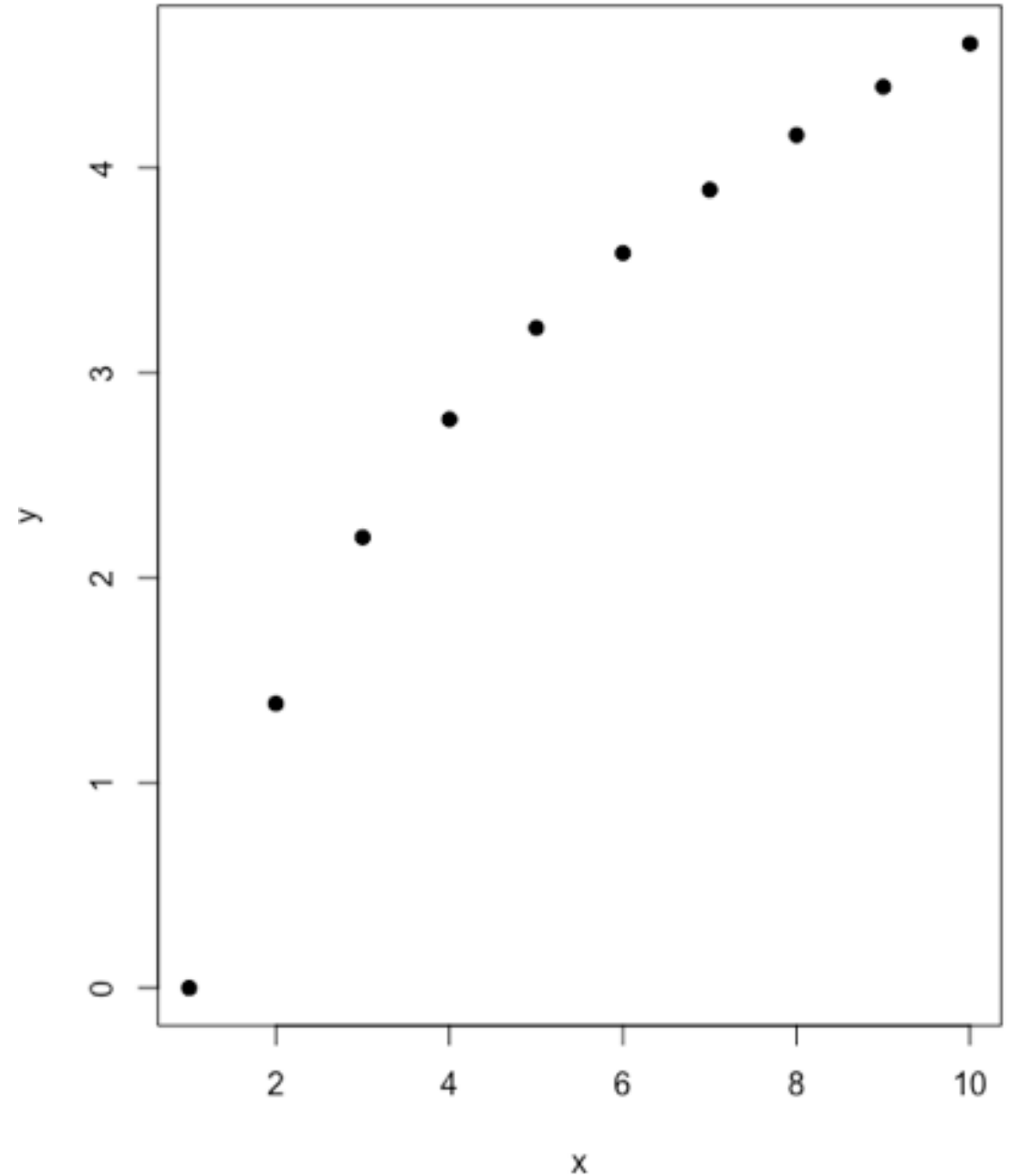
- Sparrow parental care in relationship to offspring number
- Expect a positive slope



Model selection

Which model to select?

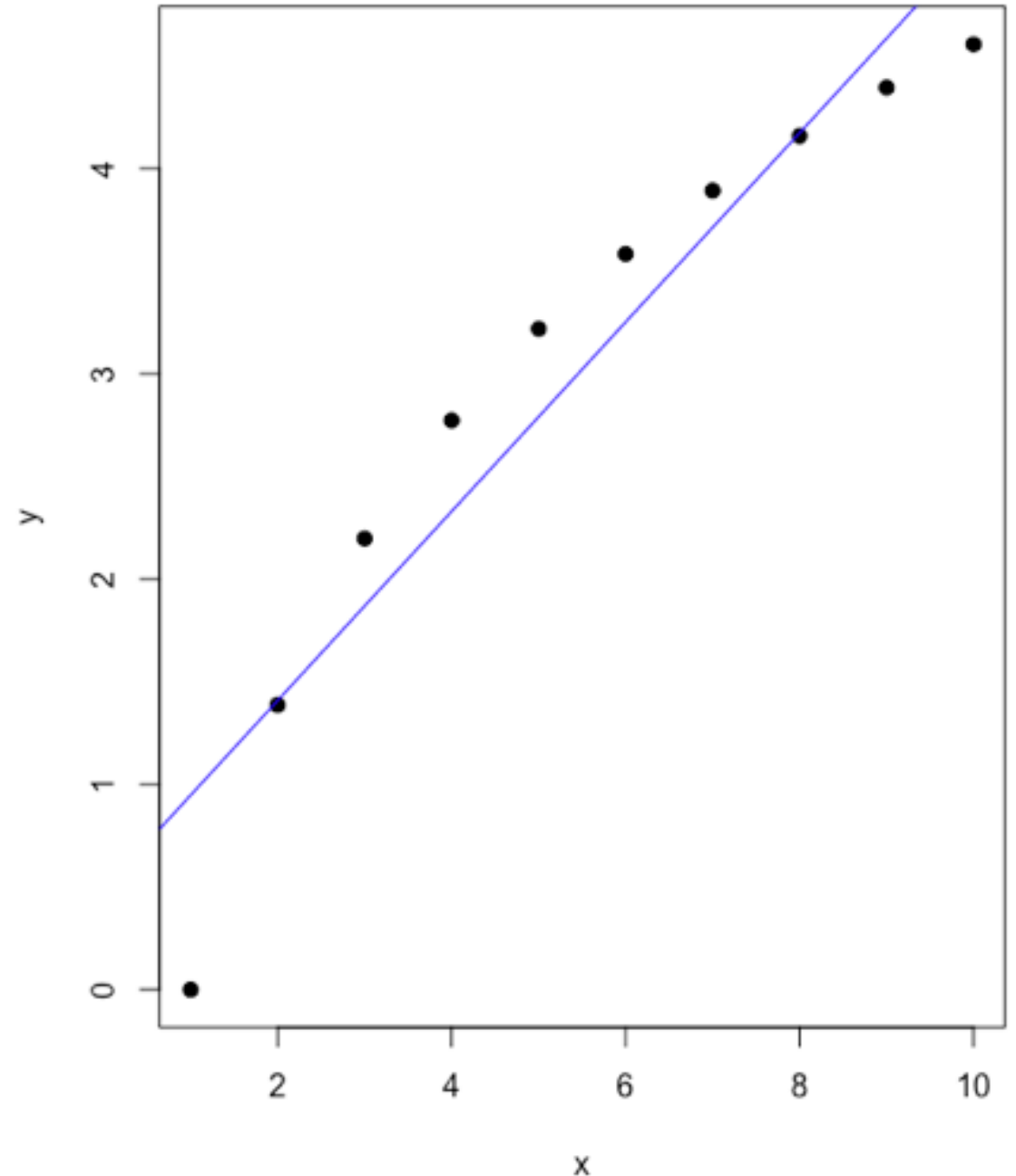
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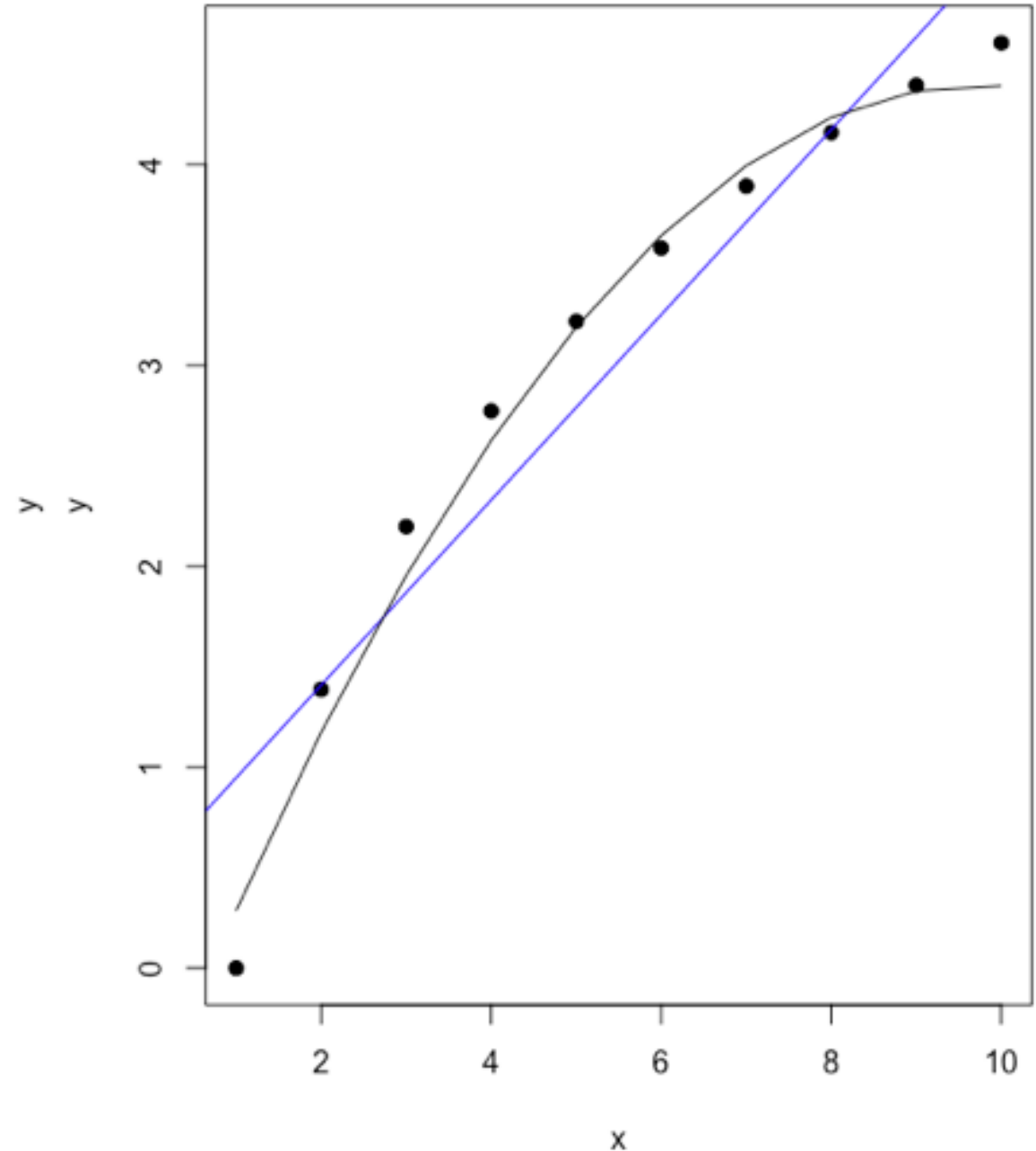
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Model selection

Which model to select?

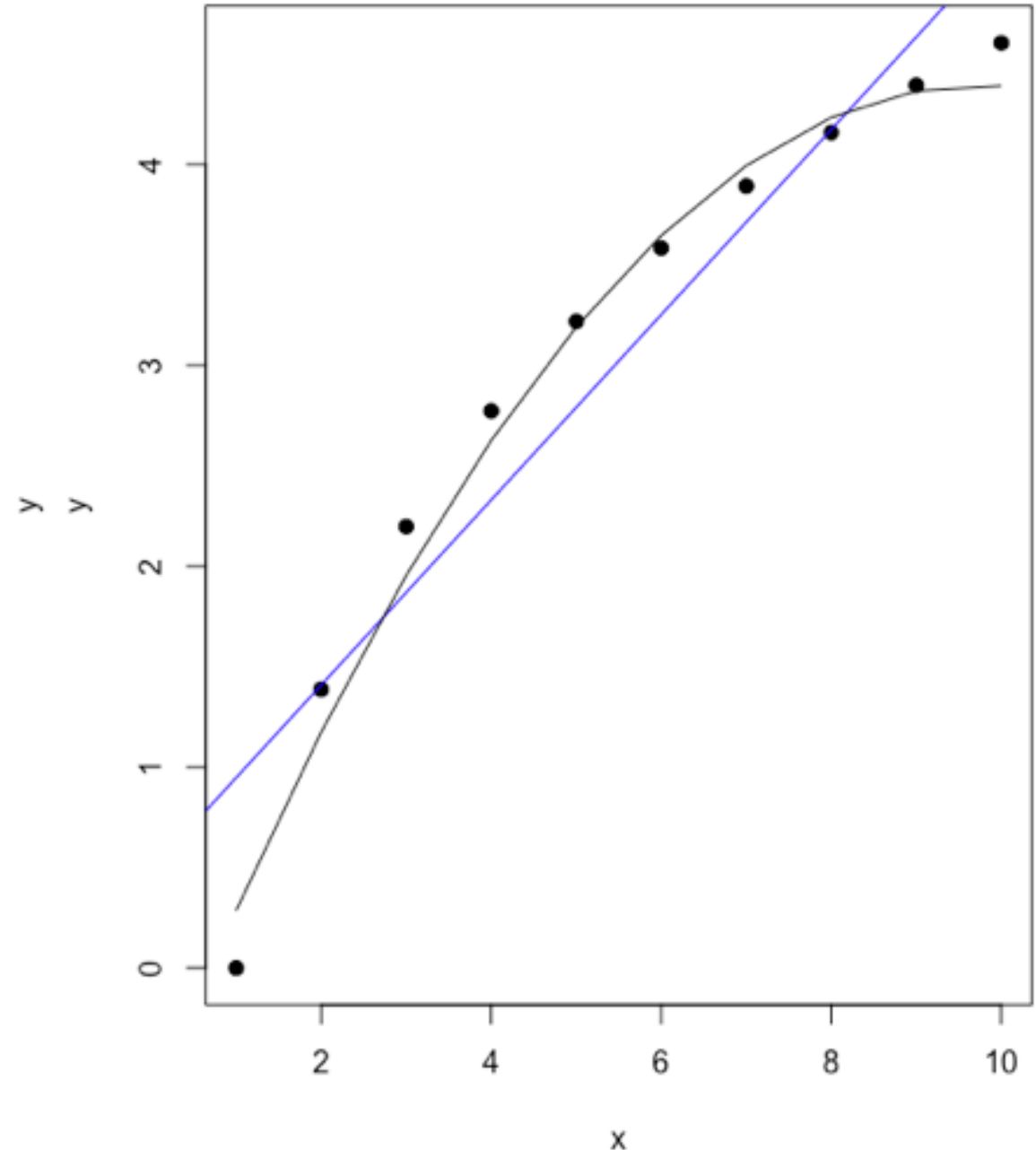
- Sparrow parental care in relationship to offspring number
- Expect a positive slope
- But diminishing



Model selection

Which model to select?

- Sparrow parental care in relationship to offspring number
- Expect a positive slope
- But diminishing
- Biology (and common sense) dictates a squared term



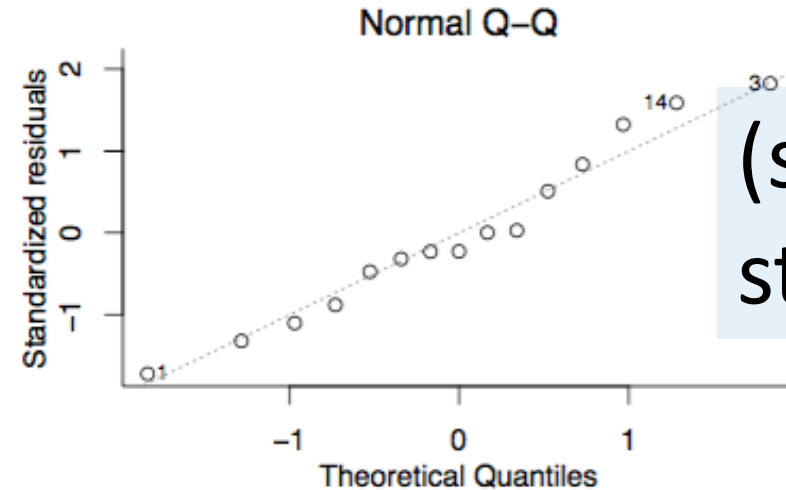
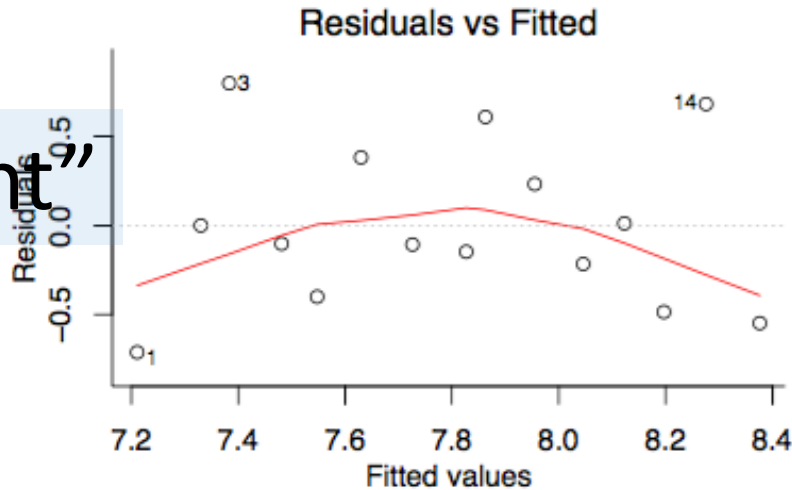
Process – common problems

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Diagnostic plots

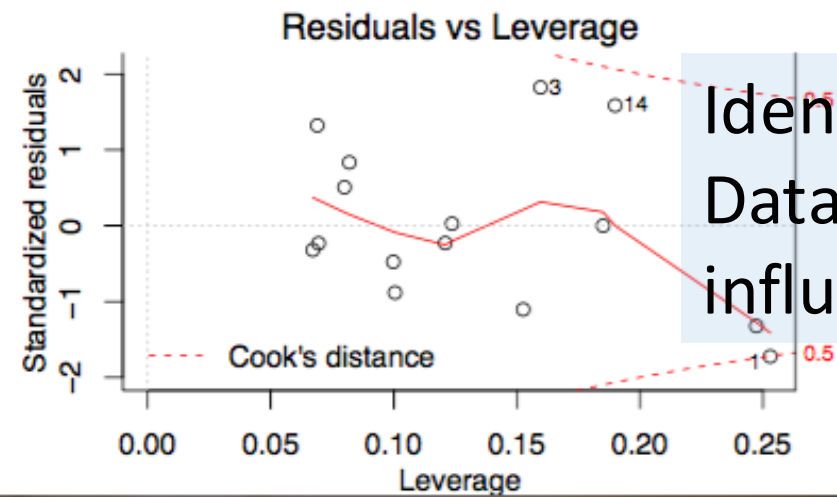
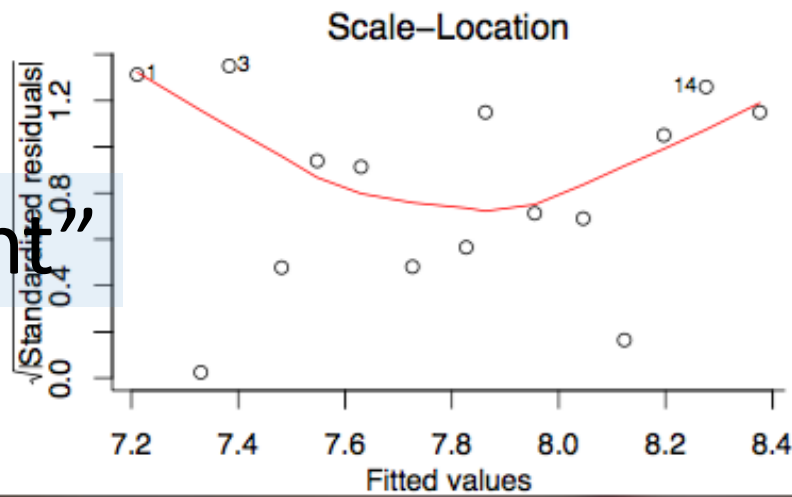
```
> mod <- lm(y ~ x, data=myData)
> plot(mod)
```

“starry night”



(somewhat)
straight line

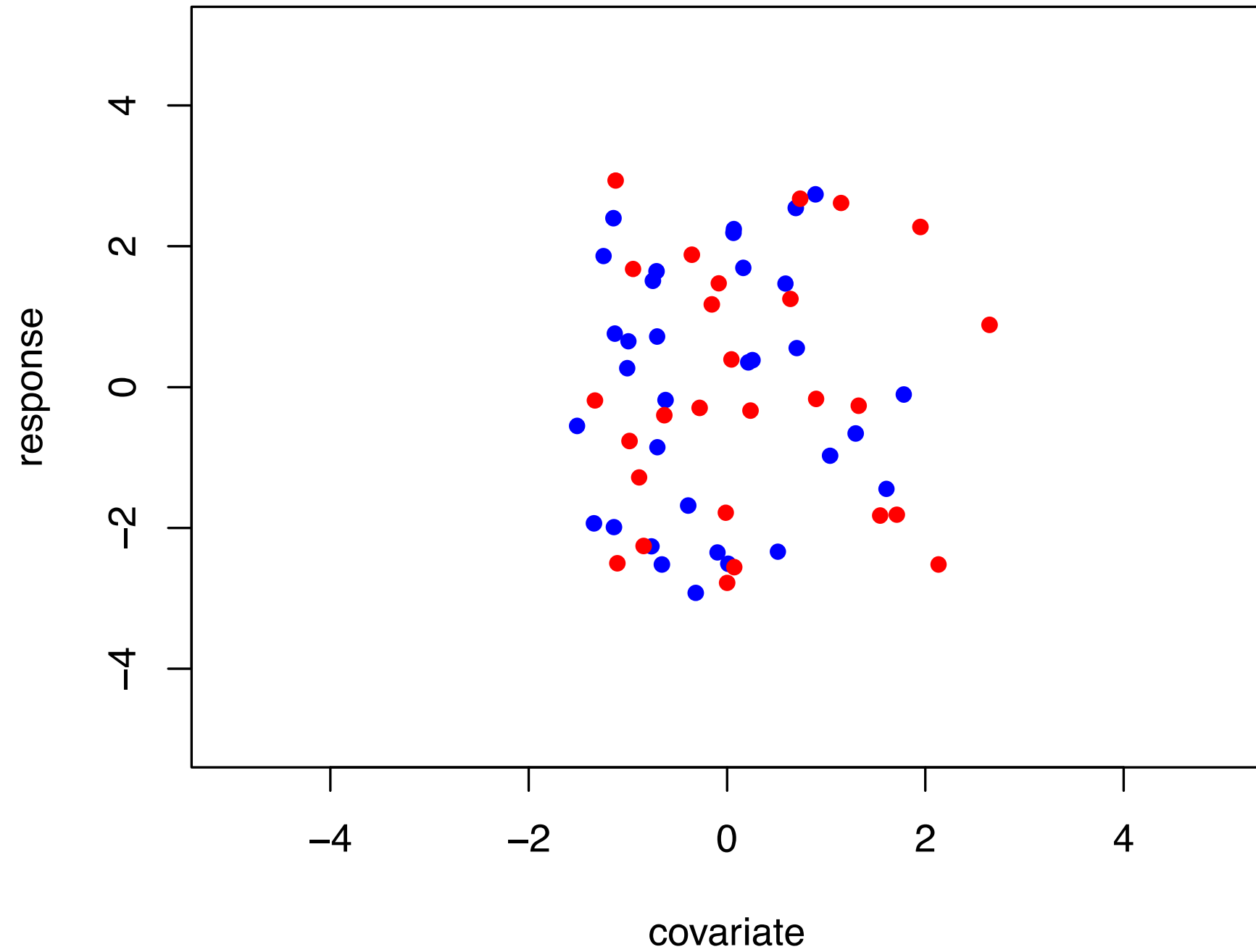
“starry night”

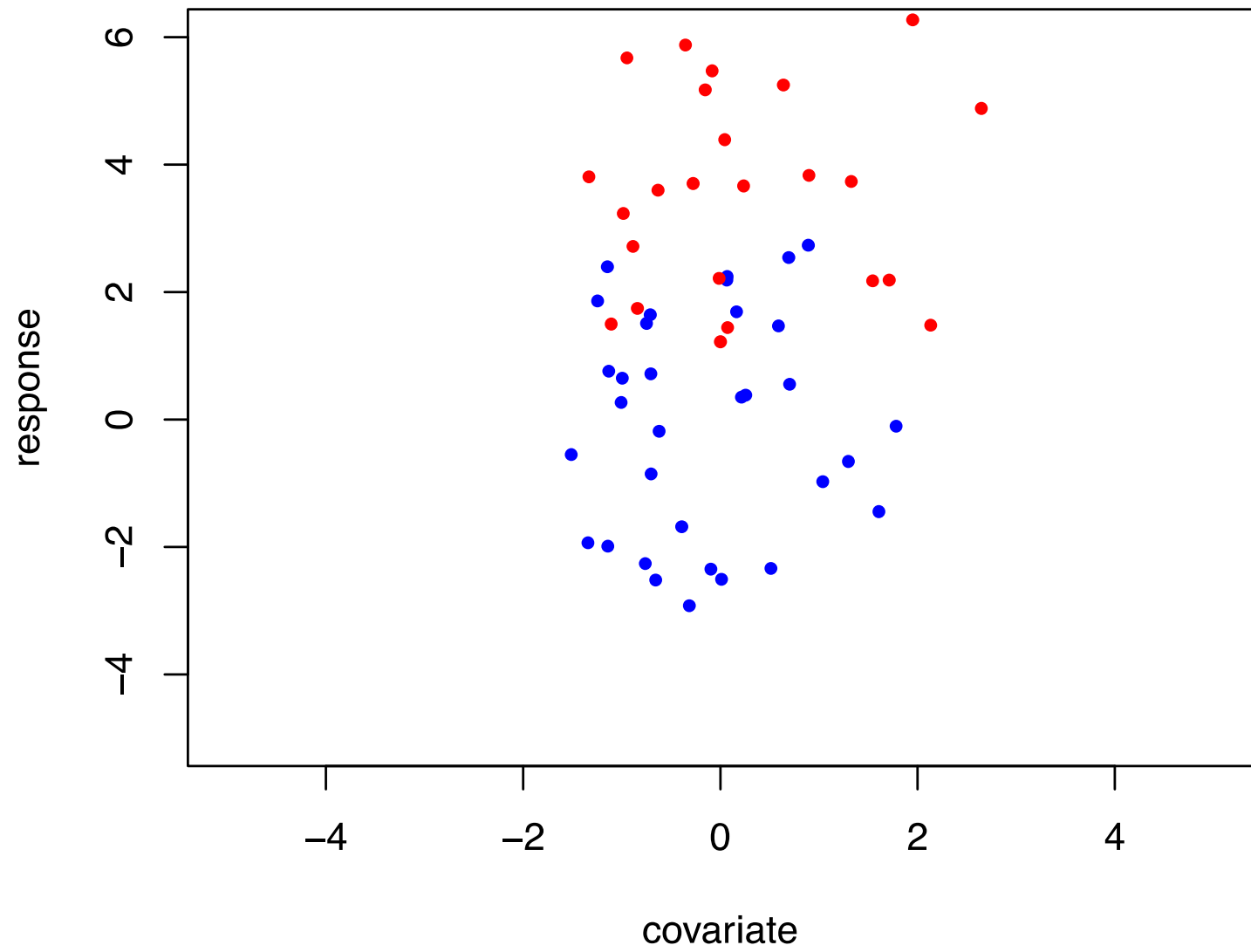


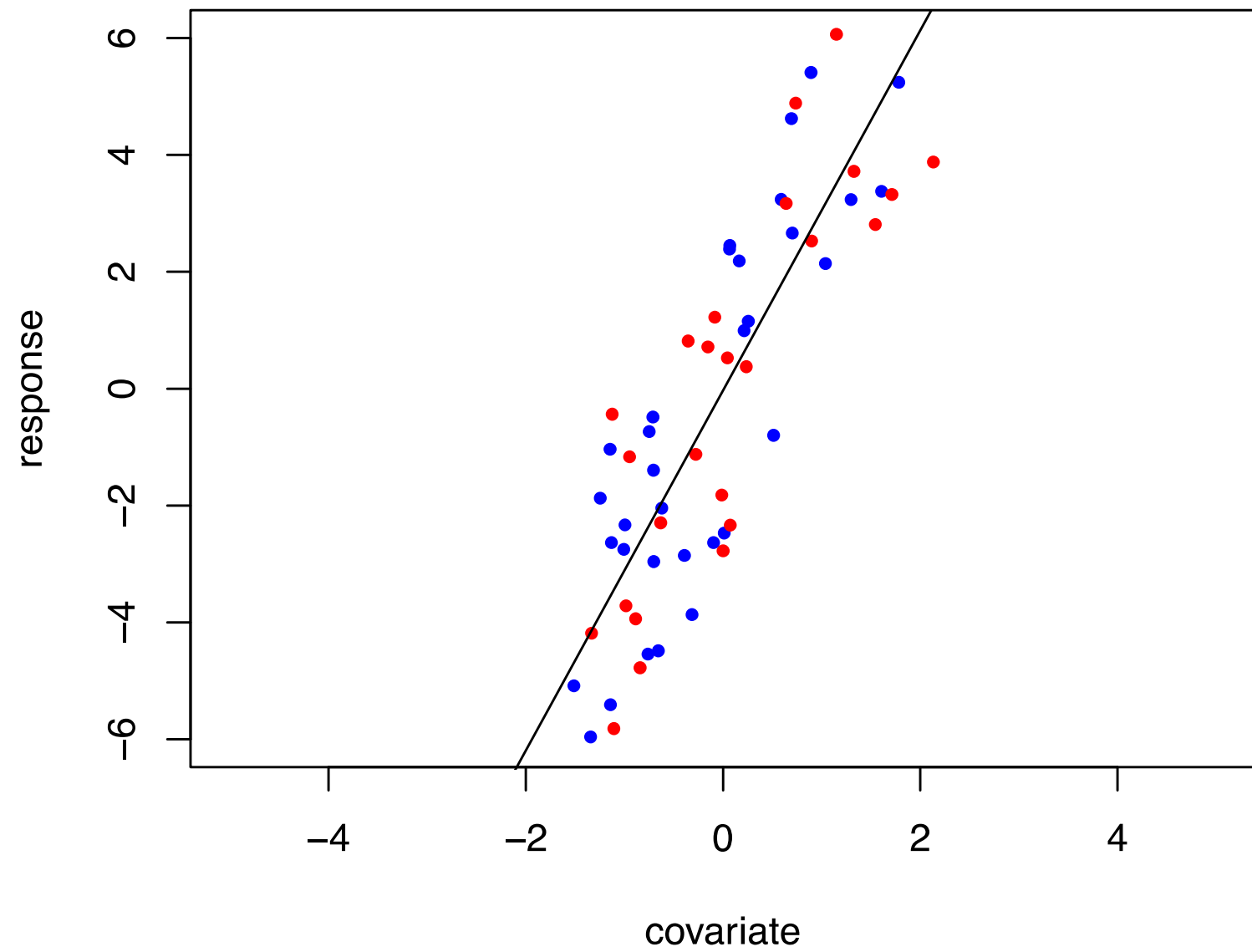
Identify outliers
Data with lots of
influence

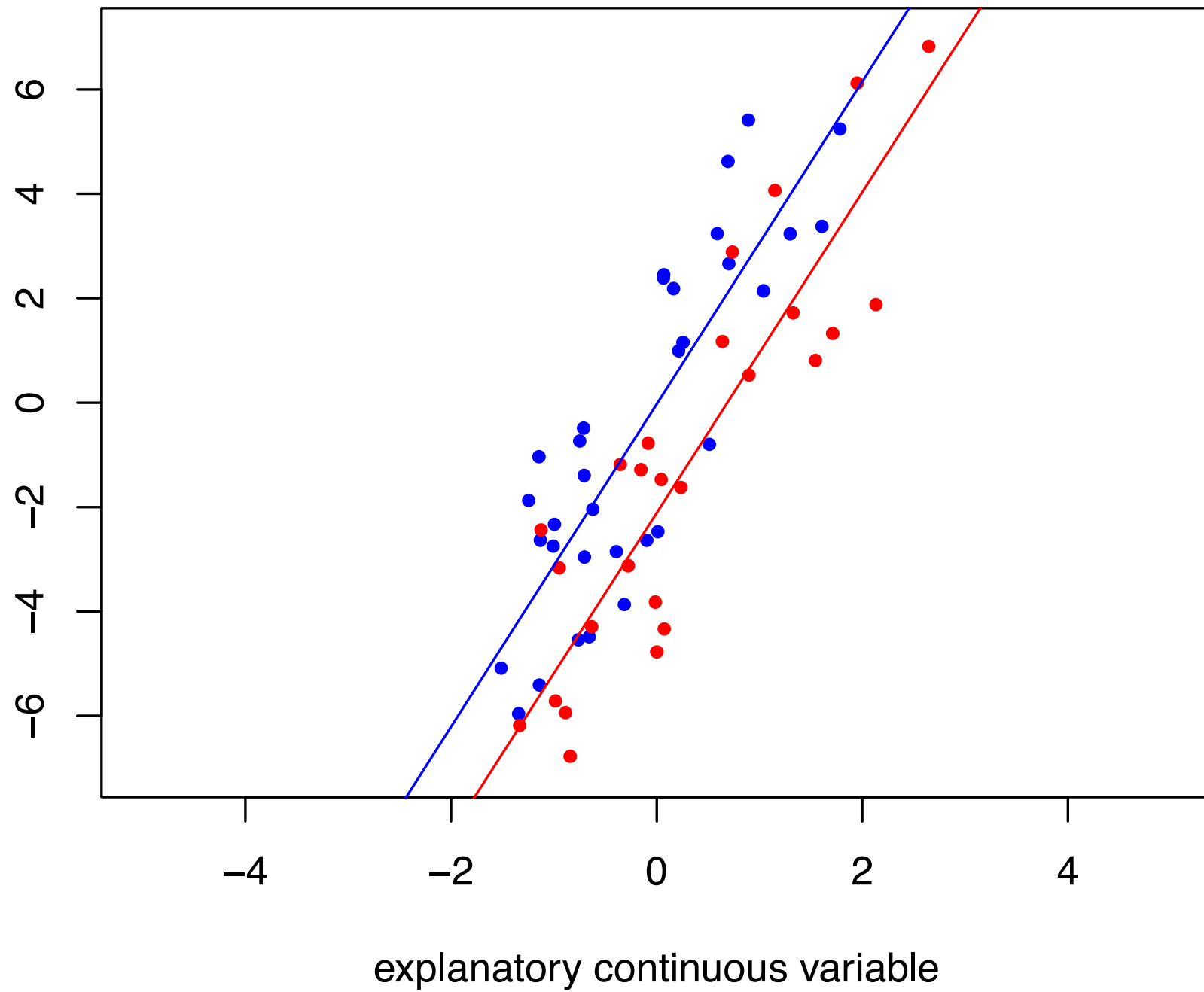
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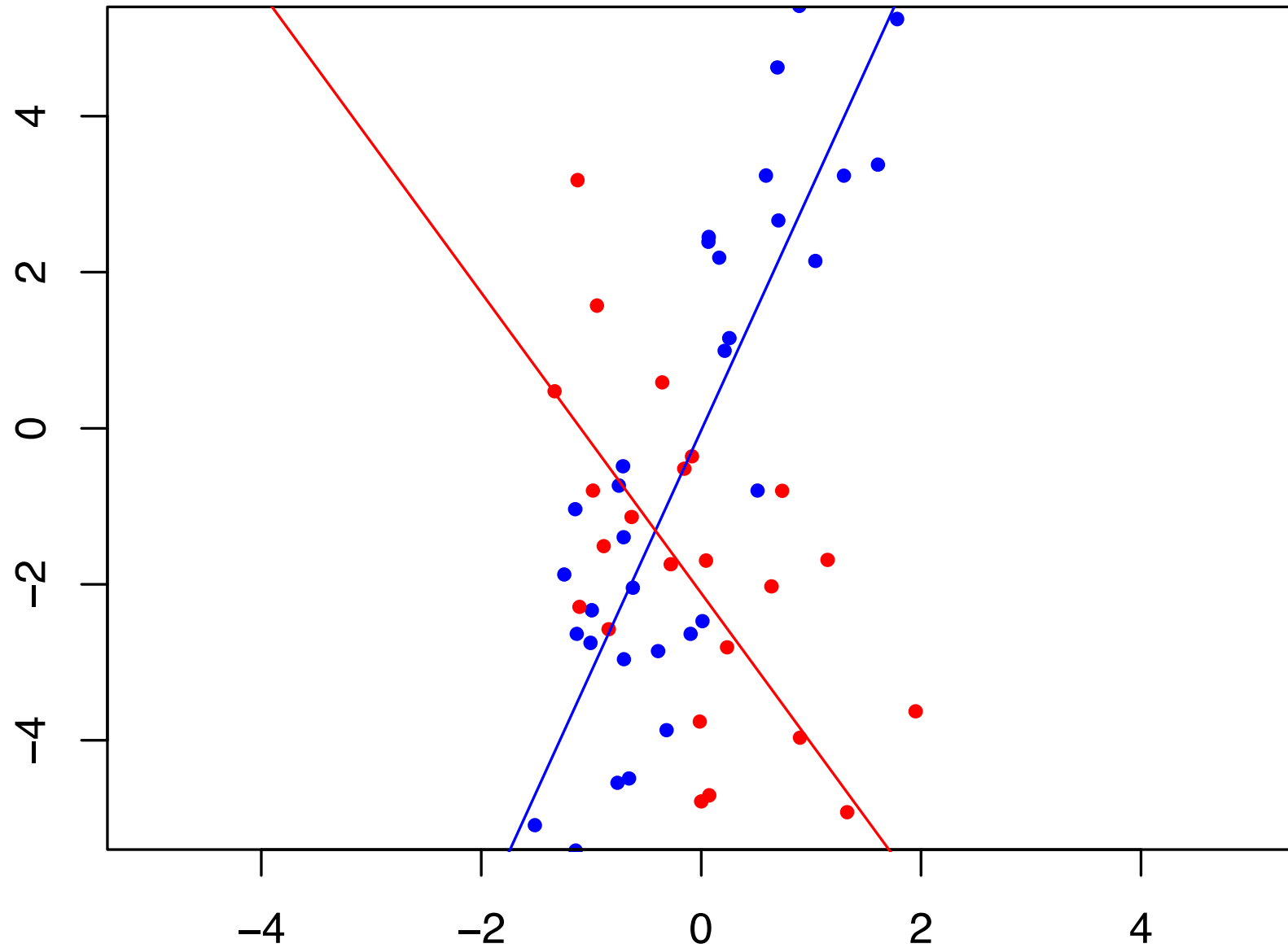








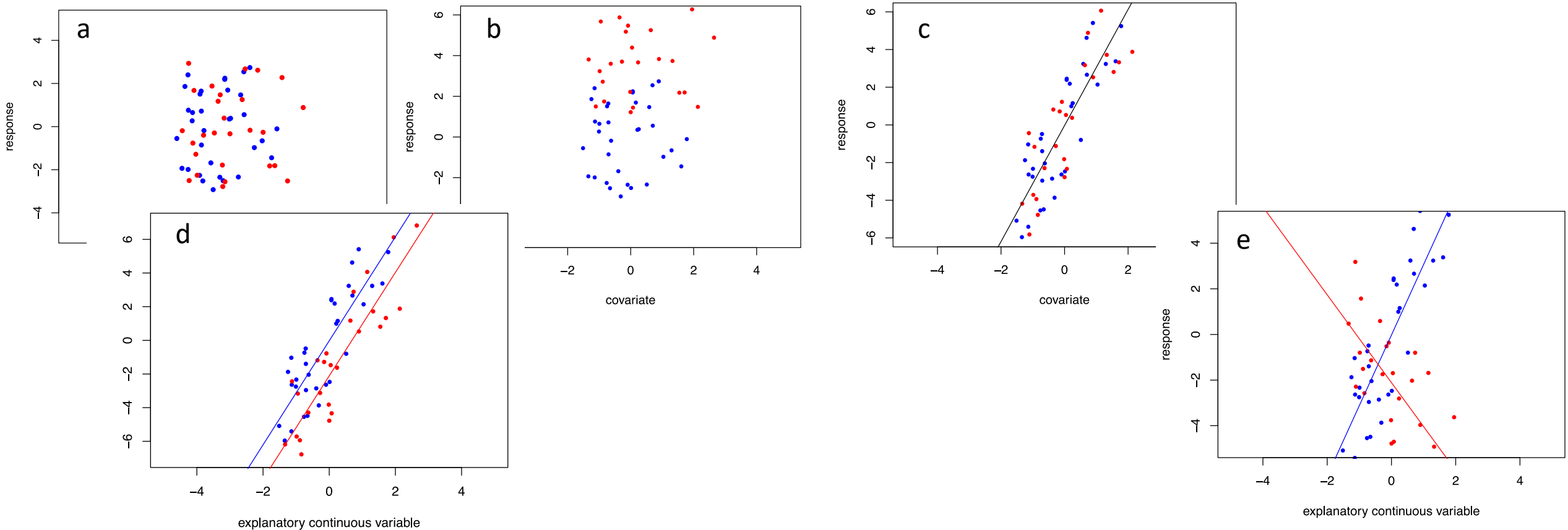
response



explanatory continuous variable

$$y_i = b_0 + b_1x_{i0} + b_2x_{i1} + b_3x_{i0}x_{i1} + \varepsilon_i$$

plot	a	b	c	d	e
Intercept					
b ₁ (sex)	0	+	0	+	+
b ₂ (tarsus)	0	0	+	+	+
b ₃ (tarsus x sex)	0	0	0	0	-



Process – common problems

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9. Model selection
10. Make a decision
11. Model validation
12. Interpretation
 - Remember units!
 - Effect sizes!